Theory of Transformer   Karl M.H. Lai

1. Ideal Transformers

Power transformer action under steady-state sinusoidal conditions can be established in terms of an *ideal transformer* embodying the transfer region. The magnetic circuit of an **ideal transformer** has **infinite permeability** and **zero reluctance**, so that a vanishingly small m.m.f. is needed to excite the magnetic flux; all the flux completely links both primary and secondary windings (there is **no magnetic leakage**); there are **no losses** in the windings or in the core.

Since the voltage $v_1(t)$ of angular frequency $\omega = 2\pi f$ is applied to the primary winding with the secondary open-circuited, the voltage is balanced by an e.m.f. induced by the rate of change of flux linkage $\psi_1 = N_1\Phi$, and the core flux must therefore vary with time.

Let $\Phi = \Phi_m \cos\ \omega t$, a variation of an angular frequency ω between oppositely directed peaks of Φ_m; then from Faraday's Law the corresponding **primary voltage** is

$$v_1(t) = \frac{d\psi_1}{dt} = \omega N_1 \Phi_m \cos\ \omega t$$

which has the **peak value** $v_{1m} = \omega N_1 \Phi_m$ and the **r.m.s. value** (as **Transformer Formula**)

$$V_1 = \frac{1}{\sqrt{2}}\omega N_1 \Phi_m = 4.44 f N_1 \Phi_m \ \rightarrow \ \text{volt per turn} = V_t = \frac{V_1}{N_1} = \frac{1}{\sqrt{2}}\omega\Phi_m = 4.44 f \Phi_m$$

Thus the primary voltage that is $V_1 = NV_t$, and that of the secondary is $V_2 = N_2 V_t$. Hence,

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$

When the secondary is open circuit, the primary input current is vanishingly small because the core is **infinitely permeable** and the flux is generated by it on **negligible m.m.f**. This means that the **winding force is zero** and there is **no infinite magnetizing inductance**. But if the secondary current is i_2, then it develops an **m.m.f.** $N_2 i_2$ acting on the magnetic circuit. To preserve unaffected the core flux Φ_m, the primary winding current i_1 must balance it such that

$$N_1 i_1 = N_2 i_2$$

at any instant. In r.m.s. terms, this becomes

$$I_1 N_1 = I_2 N_2$$

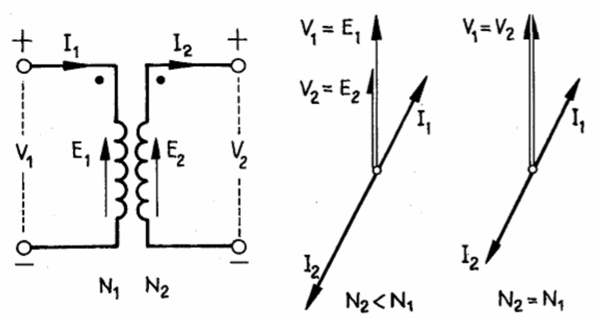


Figure 1: Ideal Transformer Model

With the 'sense' of the windings as in the figure, the primary and secondary applied phasors are coincident **in direction of length** proportional to the opposition of the current phasors. The requirement of **m.m.f. balance** is that the positive terminal, then I_1 leaves the secondary at the positive terminal, giving condition $I_1 + I_2 = 0$, following the **dot convention**. Thus a clear distinction is made between primary input and secondary output conditions.

In machine phraseology, the primary acts in a **'motor' mode**, the secondary in a **'generator' mode**. The primary input is an apparent power, $S_1 = V_1 I_1$, the secondary input is $V_2(-I_2) = -S_2$. These are conjugate:

$$S_1 = V_1 I_1 = V_2(-I_2) = -V_2 I_2 = -S_2$$

Impedance Transformation

If the secondary terminals are connected to a load of impedance Z, then $I_2 = V_2/Z$. The corresponding ratio V_1/I_1 is the **effective input impedance** of the primary:

$$Z_1 = \frac{V_1}{I_1} = (\frac{N_1}{N_2})^2(\frac{V_2}{I_2}) = (\frac{N_1}{N_2})^2 Z = k_n^2 Z$$

Summary

For **sine conditions** we have the following relations for the ideal transformer:

- **Ratios:** $V_1/V_2 = N_1/N_2 = I_2/I_1$ (infinite permeability and zero reluctance)
- **Rating:** $S = V_1 I_1$ or $V_2 I_2$ (no loss)
- **M.M.F. balance:** $N_1 I_1 + N_2 I_2 = 0$ (no flux loss)

2. Actual Transformer

The core is exposed to an actual transformer **magnetizing force**, and an established working flux requires a **magnetizing current** $i_m \neq 0$.

1. **Core excitation and m.m.f. balance**
   - The core has an effective permeability and establishes the working flux ϕ, exciting a **magnetizing m.m.f.** F_0 to be represented by the magnetizing reactance x_m such that it takes magnetizing current I_{0r} such that $N_1 I_{0r} = F_0$.
   - The m.m.f. balance changes from $F_1 + F_2 = 0$ to $F_1 + F_2 = F_0$.
   - As the core has a **non-linear B/H relation**, the magnetizing m.m.f. depends upon the **degree of core saturation**.
2. **Core conductivity and winding configuration**
   - The core is the seat of loss arising from alternating magnetization, so the exciting (primary) winding must also supply the **core-loss** to be represented by r_m, such that $P_0 = I_{0a}^2 r_m$ is the core loss.
   - The windings have individual resistances between terminals, of values r_1 and r_2 determined by the **conductor resistivity** and augmented by **eddy current loss**.
3. **Leakage inductance and imperfect coupling**
   - The windings are not perfectly coupled, so that not all the flux developed by the primary links the secondary, and the latter produces further flux not linking the primary.
   - As the leakage fluxes x_1 and x_2 are established in paths partly *external to the core*, the **leakage inductances** can be taken as constant. At a **given operating frequency** they can be represented in an equivalent circuit by leakage reactance.

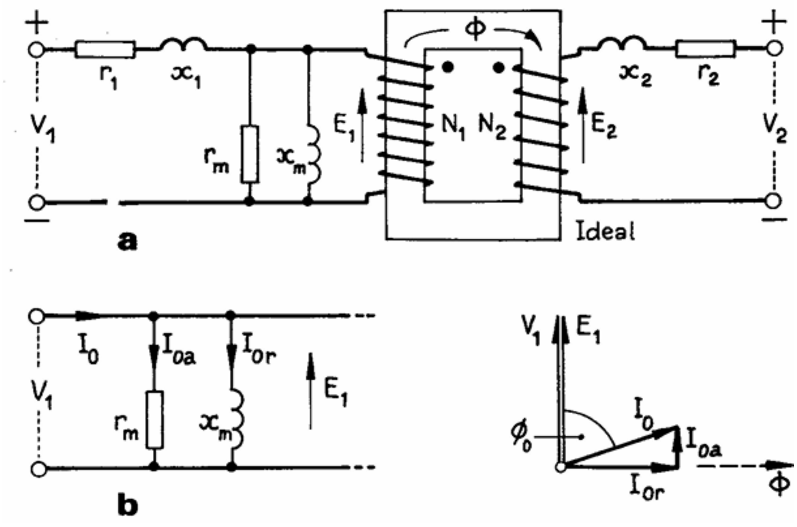


Figure 2: Transformer on No Load

Open Circuit Condition:

Let the ideal-transformer part of the system carry a sinusoidally alternating mutual (or working) magnetic flux of peak value $\Phi_m$and angular frequency $\omega = 2\pi f$. By definition this flux varies with time, inducing therein the emf:

$$e_t = \frac{d\Phi}{dt}$$

having the peak value $\omega\Phi_m$and the r.m.s. value:

$$E_t = \frac{\omega\Phi_m}{\sqrt{2}} = \sqrt{2}\pi f\Phi_m = 4.44 f\Phi_m$$

The total primary m.m.f. $= N_1 I_1$turns in series is $E_1 = N_1 E_t$, and for the secondary correspondingly $E_2 = N_2 E_t$. We shall first consider the secondary terminals to be open-circuited: this is the **no-load condition**.

The actual core requires the magnetizing m.m.f. F_0, which in equivalent-circuit terms is the current in x_m, namely I_0. Further, the core loss is provided in the equivalent circuit by the current I_{a0}, in parallel. The emf E_1balances the primary applied voltage V_1(neglecting the very small volt drops due to the magnetizing and loss currents flowing through the primary leakage impedance z_1). Thus:

$$E_1 = V_1$$

and the **no-load primary input current** is:

$$I_0 = \sqrt{I_0^2 + I_{a0}^2}$$

The no-load power is therefore almost identical with $V_1 I_{0a} = P_0$, which is the core-loss power.

Short Circuit Condition:

Suppose the secondary terminals to be short-circuited, as in **Fig. 3**. The secondary e.m.f. $E_2$is absorbed entirely in circulating I_2through the leakage impedance $z_2 = r_2 + jx_2$, and there is no load to serve.

To limit $I_2$to the level of normal rated current it will be necessary to limit the primary voltage $V_1$to about 0.1 or less of normal. The primary voltage is employed in providing $E_1$and the volt drop in the primary leakage impedance z_1. As $z_1 \approx z_2$, it follows that $E_1$is about one-half of V_1.

With V_1 equal to 0.1 p.u. then E_1 is about 0.05 p.u., which means that the mutual flux Φ_m is of the same order. The **core loss**, which is proportional approximately to the **square of the flux density**, is consequently about 0.0025 of what it is under full-load.

In the equivalent circuit it is legitimate to ignore completely the shunt magnetization branches r_m and x_m, to give the simplified equivalent circuit of **Fig. 3(b)** and the corresponding phasor diagram shown in **Fig. 3(c)**. The assumptions made in deriving these figures are, it is seen, $E_1 \approx V_1$ and $E_2 \approx V_2$, and the phasor diagrams show somewhat awkward. We therefore adopt the **unity turns-ratio equivalent**.

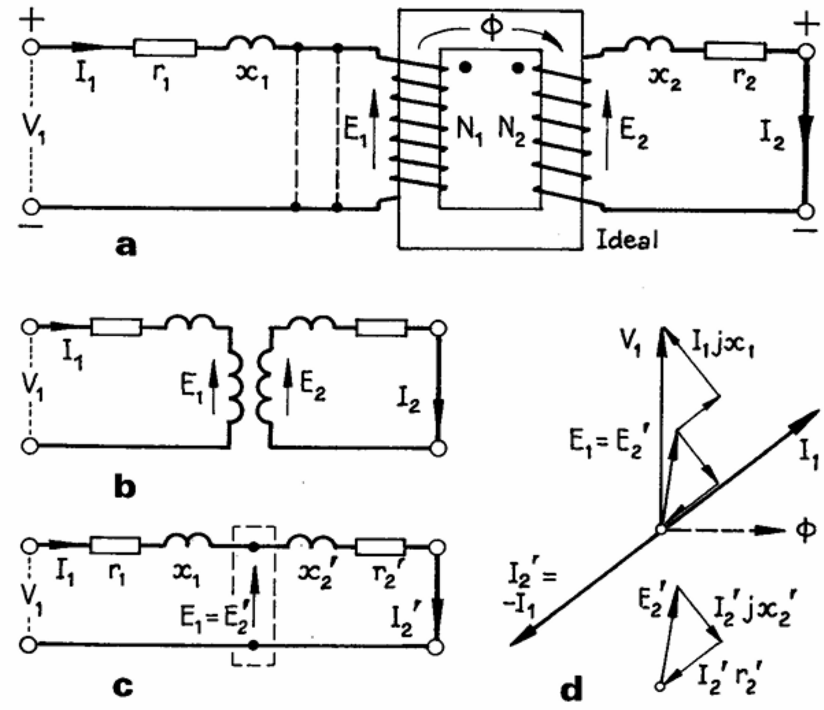


Figure 3: Transformer on Short Circuit

Suppose that $N_1 = N_2$ and that we consistently take $E_1 = E_2$ in the ideal-transformer part of the equivalent circuit. Energy is now transferred from primary to secondary without any change of voltage or current, so that there is no need to have an ideal transformer at all. To retain such simplicity when $N_1 \neq N_2$, it is necessary to imagine the actual secondary winding of N_2 turns to be replaced by an equivalent winding of N_1 turns. There is then **no change in the turns-ratio** (**loss** nor in the **effective leakage reactance**).

The transformed secondary must have a resistance r_2' and leakage reactance x_2' such that:

$$r_2' = r_2(\frac{N_1}{N_2})^2,\ x_2' = x_2(\frac{N_1}{N_2})^2 \rightarrow I_2'^2 r_2' = I_2^2 r_2, I_2'^2 x_2' = I_2^2 x_2$$

Hence the resistance and leakage reactance of the equivalent secondary are:

$$r_2' = r_2 k_n^2, x_2' = x_2 k_n^2, \qquad k_n = \frac{N_1}{N_2}$$

Transformer On-Load Condition:

If the load impedance on **Z** is so chosen that a partially inductive secondary winding is formed as in **Fig. 3**, then the secondary current $\mathbf{I_2}$ will produce an output current to maintain the working flux and the balance of $\mathbf{N_2I_2}$ on the magnetic circuit. To maintain the working flux $\boldsymbol{\phi_m}$ and the balance of $\mathbf{N_1I_1}$, the primary must take an input current $\mathbf{I_1}$ to develop the ampere turns required. The primary, which must always be in balance with the applied voltage $\mathbf{E_1}$, at the same time must supply a current to provide for the magnetizing component $\mathbf{I_m}$ and balance $\mathbf{F_0}$, and at the same time to provide in the primary the component $\mathbf{I_1}$ to balance $\mathbf{F_2}$. The secondary current $\mathbf{I_2}$ must result in a net magnetizing m.m.f. $\mathbf{F_0}$. Thus a secondary current causes a counter-flux to flow in the magnetizing and the common magnetic core m.m.f. to result in net magnetizing m.m.f. $\mathbf{F_0}$.

If we adopt the more easily drawn condition with a **partially inductive** secondary current, the phasor currents can be used to represent as is shown for a $\mathbf{N_2I_2}$ primary and secondary load in **Fig. 4**.

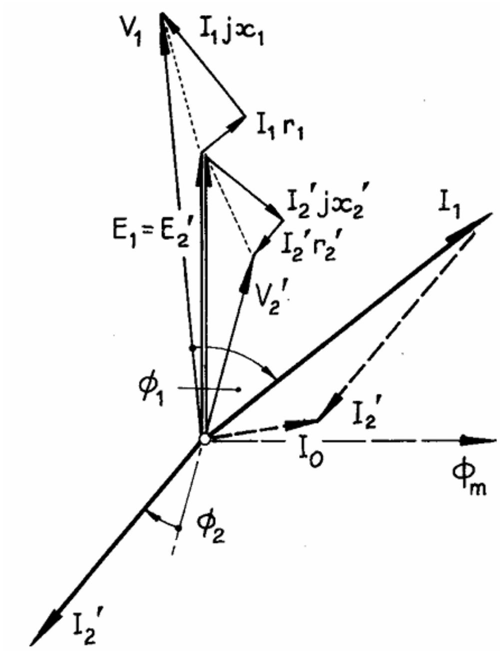


Figure 4: Transformer On Load Condition – Phasor Diagram

3. Losses and Efficiency

The transformer has no moving parts, and its efficiency has far less significance than its losses, which are:

- **Core Loss:** resulting from the alternating core magnetization.
- **Dielectric Loss:** in insulating materials, particularly in the oil and solid insulation of high-voltage units.
- **I^2R Loss:** in the windings.
- **Load (Stray) Loss:** largely the result of leakage magnetic fields inducing eddy currents in the tank walls and the conductors.

A close approximation to the losses can be obtained from **open-** and **short-circuit tests**, without operating the transformer on load.

No-Load Loss

With open-circuited secondary, the primary current is I_0. The I^2R loss due to it is in most cases quite negligible (e.g. 1/400 of the rated normal I^2R loss). The **no-load power input** is consequently accounted for with **core** and **dielectric loss**, the latter having significance only for transformers operating at very high voltages. The e.m.f. E_1 is substantially identical with V_1, the conditions being represented by the equivalent circuit in respect of the **core magnetization**. If the **dielectric effect** is negligible, the no-load input

$$P_0 = V_1 I_0 \cos\theta_0 = V_1 I_{0a}$$

is the loss in the core.

In large HV transformers, however, it will also include the **dielectric loss** P_d in the dielectric material. The core magnetizing current I_{0m} is accompanied by a **leading capacitive current** I_{0d}, the effect of which is to reduce the lagging reactive component I_{0r} of the no load current. Thus, the core and dielectric losses (generally assumed constant and independent of load) are determined by means of **open circuit test** at nominal voltage and frequency.

Short Circuit Loss

If rated secondary current is circulated in the **short-circuit test** by means of a low value of a low voltage of primary voltage (short frequency), the circuit primary is

$$V_1 = I_1\big((r_1 + r_2') + j(x_1 + x_2')\big) = I_1(R_1 + jX_1) = I_1 Z_{eq}$$

and the power input $P_{in} = I_1^2 R_1$, which is equal to short-circuit loss in the primary and secondary windings.

In practice, the active power input includes (stray) load loss, for with normal primary and secondary currents the **leakage fluxs** are normal, producing **eddy current loss** in the metallic parts of the transformers adjacent to the windings (e.g. shell and core). Thus a **short-circuit test** at rated current and frequency can establish the **load loss** to a close approximation. It also provides information about the **leakage impedance** and **effective resistance**, from which the leakage reactance can be calculated.

Power Efficiency

The peak flux normally varies so little between **no-load** and **full-load conditions** that the core loss can be considered as constant at the **no-load value** P_i. If the short-circuit loss at rated current is P_c, the loss at any fraction $k$of the rated load S is k^2P_c. The total loss at load kS of power factor $\cos\phi$ is $P_i + k^2P_c$, and the output/input power ratio, i.e. the power efficiency is

$$\eta = \frac{kS\cos\,\phi}{kS\,\cos\,\phi + P_i + k^2P_c} = 1 - \frac{P_i + k^2P_c}{kS\cos\phi + P_i + k^2P_c}$$

Solving $d\eta/dk = 0$, it gives

$$k = \sqrt{P_i/P_c} \rightarrow \; P_i = k^2P_c$$

for the maximum efficiency condition with a load of fraction k of full load rating.

4. Regulation

The **regulation** is the fall (or rise) of secondary terminal voltage between no-load and full-load conditions with the primary voltage at constant rated value. It is usually quoted as a **per-unit value** for a specified power factor.

On no load the secondary terminal voltage is very nearly

$$V_2' = V_1$$

On full load corresponding to a primary current I_1 it is given by the phasor expression based on **Fig. 5(a)**:

$$V_2' = V_1 - I_1(z_1 + z_2') = V_1 - I_1 Z_1$$

ignoring magnetization current. The regulation is the scalar difference between V_1 and V_2', and as shown in **Fig. 5(b)** it depends on the power factor of the secondary load.

It is convenient to define the per-unit resistance (or per-unit full-load I^2R loss) and the per-unit reactance respectively as:

$$\varepsilon_r = \frac{I_1^2 R_1}{V_1 I_1} = \frac{I_1^2 R_2}{V_2 I_2} = \frac{P^c}{S}$$

$$\varepsilon_x = \frac{I_1^2 X_1}{V_1 I_1} = \frac{I_2^2 X_2}{V_2 I_2} = \frac{I_1 X_1}{V_1} = \frac{I_2 X_2}{V_2}$$

These can be combined to give the ***per-unit leakage impedance***

$$\varepsilon_z = \sqrt{\varepsilon_r^2 + \varepsilon_x^2}$$

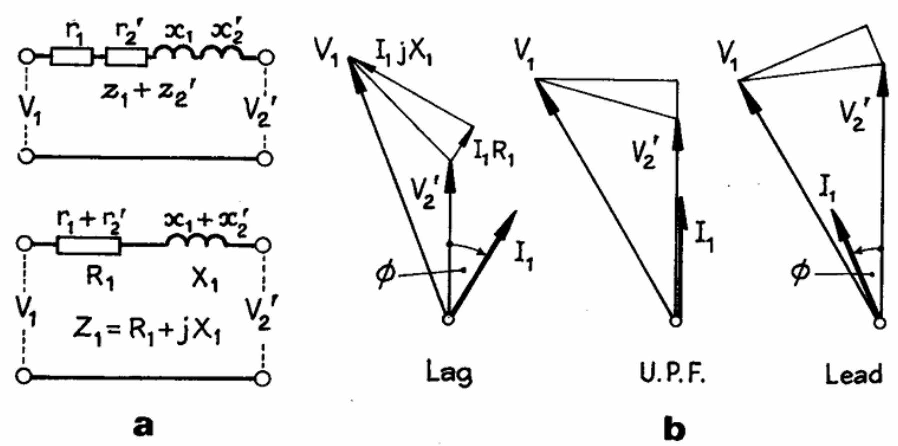


Figure 5: Voltage Regulation

The **per unit regulation** can be represented by the scalar equation,

$$\varepsilon = V_1 - V_2' = 1 - V_2'$$

$$\varepsilon = \varepsilon_r \cos\phi + \varepsilon_x \sin\phi + \frac{1}{2}(\varepsilon_r \cos\phi - \varepsilon_x \sin\phi)^2 + \cdots$$

If ε_z is less than 0.1 (or 10%) the squared term can be ignored to give:

$$\varepsilon = \varepsilon_r \cos\phi + \varepsilon_x \sin\phi$$

The **full-load regulation** is a maximum when

$$\phi = \tan^{-1}(\frac{\varepsilon_x}{\varepsilon_r}) = \tan^{-1}(\frac{X_1}{R_1})$$

this is a moderately low lagging power factor. It is zero for

$$\phi = -\tan^{-1}(\frac{\varepsilon_x}{\varepsilon_r})$$

corresponding to a **high leading power factor**, and for **lower leading power factors** the secondary voltage rises between no-load and full-load condition. A typical case for $\varepsilon_r = 0.01$ and $\varepsilon_x = 0.04$ is shown in **Fig. 7**.

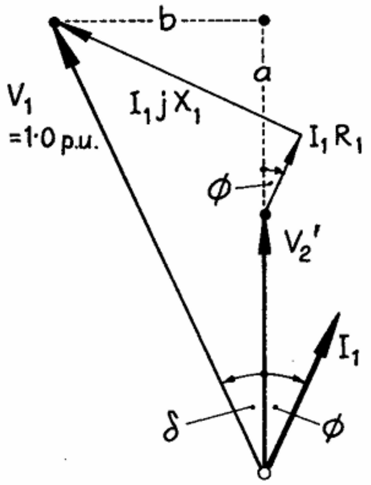


Figure 6: Assessment of Regulation

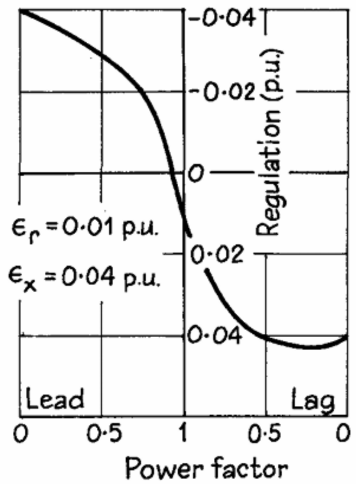


Figure 7: Regulation Curve

5. Harmonics

A typical magnetizing current waveform for sinusoidally varying flux density contains strong harmonics, especially the third harmonic. Consider the following cases:

In **Fig. 8(a)**, a flux density consisting only of the fundamental, with peak B_{1m}, requires an **exciting current** with 1st and 3rd harmonics.

In **Fig. 8(b)**, a purely sine magnetizing current results in a flux density B distorted by **saturation**, introducing a 3rd harmonic flux density B_3.

Neglecting **hysteresis phase-shift**, the flux density can be taken as the sum of a fundamental $B_{1m}\cos\ \omega t$ and a 3rd harmonic $B_{3m}\cos\ 3\omega t$, where the emf per turn embracing a net core area A_i is

$$e = \omega A_i[B_{1m}\cos\ \omega t + 3B_{3m}\cos\ 3\omega t]$$

The consequences of this waveform will be examined, and Fig. 8(b) shows a peaked waveform.

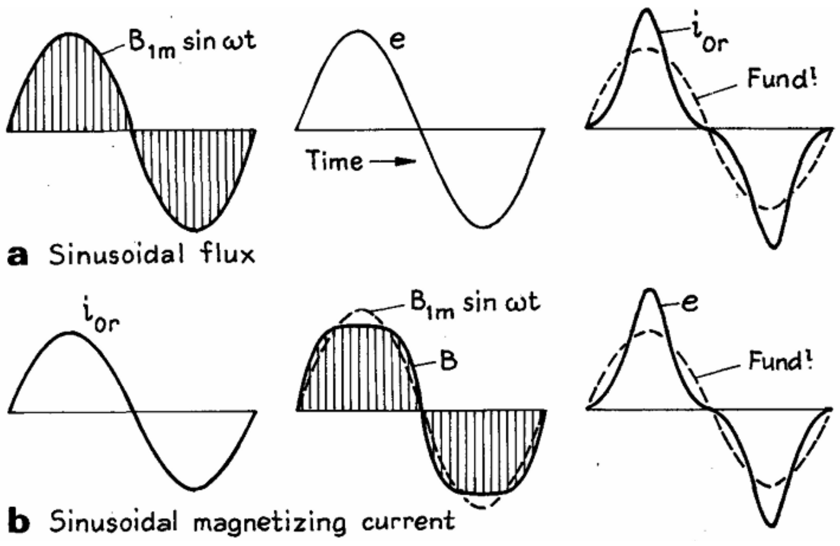


Figure 8: Flux, EMF and Magnetizing Current Harmonics

Single-phase Transformer

A sinusoidal supply voltage cannot give rise to harmonic magnetizing currents directly; it supplies **fundamental reactive power**, part of which is converted to **harmonic reactive power** in the core with its nonlinear **B/H characteristic**.

The flow of **3rd harmonics current** of a transformer implies the presence of a **3rd harmonic m.m.f.** and therefore a **corresponding harmonic in flux waveform**. The harmonic currents flow through the **effective 3rd harmonic impedances** of the **primary winding** and **its supply network** as shown in **Fig. 8**, and (to a limited extent) in the secondary winding and its load.

These flux harmonics have a waveform intermediate between the two shapes in **Fig. 8**, being more flattened (and the magnetic current peaks less peaked) if the 3rd harmonic impedance is high.

If the **network impedance** is negligible, the amplitude of the 3rd harmonic m.m.f. is experienced in driving the harmonic current through the appropriate **triplen-frequency impedance** of the transformer and no harmonic voltage appears at the terminals of the primary; but if not, i.e. the network impedance is large, the voltage waveform is distorted.

Three-phase Banked Single-phase Transformers

Each of the three separated cores must carry the flux demanded by its **operating conditions**, but the effects are now modified by the **phase interlinkage**. In the following, both supply and load are balanced and star connected.

(a) Dd Connection.

- In a symmetrical three-phase voltage, the fundamental and the harmonics of order (3n$\pm$1), where n = 1, 2, 3, ... are all represented by phasors with a 120° displacement.
- The **triplen harmonics** are **voltage co-phasal** in all three phase windings and peak simultaneously.
- **Delta-connection** provides a **closed path** to triplen EMF which **circulates** triplen currents in the closed mesh and are absorbed by the **triplen leakage impedance drop**.
- The supply voltage provides only fundamental magnetizing current, excites a **flattened flux** waveform with triplen content and induces **triplen EMF**. The latter triplen current circulates in the delta, to add to the sinusoidal supply current and to restore flux waveform nearly to pure sine wave. Hence, primary voltage and current are sinusoids, the flux is nearly so, and the necessary triplen magnetizing current is circulated around the closed mesh.

(b) Yd and Dy connection without Neutral.

- Either connection operates as in (a), expect that with a closed mesh on one side only. The triplen impedance is greater and the flux waveform diverges rather more from the sinusoidal.

(c) Yy Connection without Neutral.

- The triplen EMF are all directed **simultaneously away from** or **towards the star points** and therefore cancel between any pair of lines. Consequently, no triplen current can flow, and the input magnetizing current are sinusoidal.
- The **flux waveform is flattened**, and its **triplen harmonics providing the triplen EMFs**.
- Balance between lines of the supply voltage and the EMF is maintained, but the effect on star point voltage is to make it exhibits the oscillating neutral phenomenon as shown in **Fig. 9**. The fundamental EMF $e_{r1}.e_{y1}$ and e_{b1} are mutually displaced by $2\pi/3$ rad. The superposed 3rd

harmonics e_3 is co-phasal in all three legs, and the resultant phase voltages are shown with the dotted lines. The diagram shows four successive instants displaced by 1/12 periods.

- All the line-to-star-point voltages fluctuate, causing "**neutral oscillation**".

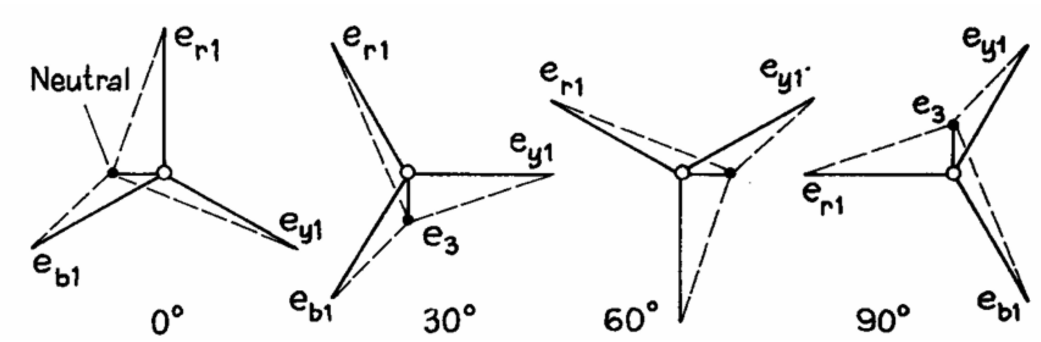


Figure 9: Oscillating Neutral

(d) Yy connection with Neutrals

- The presence of neutral connection effectively separates the three 1-ph transformers, the neutral carrying the co-phasal triplen compensating current.

(e) Other Yd connections

- A **delta connection** always provides a closed path for triplen currents
- A **neutral connection** to the supply allows triplen currents to flow into the supply network.

(f) Stabilizing D (Tertiary) Winding

- An **auxiliary winding** on each transformer, connected in delta, and applied to Yy or Yz connected 1-ph banks to provide a **closed path** in which **triplen current can circulate**. In effect, it decreases the **zero-sequence impedance**.

Three-phase Unit Transformer

The triplen flux harmonics are significant in a 3-phase unit transformer only if the phases are **magnetically interlinked**.

- **3 Phase-3 Limbs Shell-type transformers** have magnetically separate phases, so triplen harmonics are not a concern.
- **3 Phase-3 Limbs Core-type transformers**, which are more common, have magnetically interlinked phases. Triplen fluxes in these designs must return through external paths with high reluctance, potentially causing issues like **eddy-current losses** in the tank.
- **Five-limbed cores** mitigate these problems by offering internal magnetic paths for triplen fluxes via the end limbs.

Harmonic Current Compensation

For core densities exceeding 1.5T, the harmonic current content markedly rises. It may in some cases be reduced by **special compensating arrangements**.

- A saturated Yy isolated-star-point **core-type transformer** may have magnetizing current waveform as shown in **Fig. 10(a)**. The 3rd harmonics is precluded by the connection. The **high reluctance** to 3rd harmonics fluxes forces the **flux towards sinusoidal**.
- If the transformer has a **5-limb core** in **Fig. 10(e)**, a 3rd harmonics flux path is provided by the end limbs, and the flux waveform flattens.
- The magnetizing current waveform in **Fig 10(b)** shows that the 5th harmonic is reversed and the fundamental slightly reduces.
- Comparison of **Fig. 10(a)** and **(b)** suggests that if the end limbs had a suitable reluctance, there would be zero 5th harmonics in the magnetizing current as in **Fig. 10(c)**. The method applies only to isolated Yy and to Yz connections.
- The **parallel connection of star/star with a delta/delta transformer** result in suppression of 5th and 7th harmonics current since these are equal and opposite in phase, cancelling in the combined input.
- The same effect in a single transformer may be obtained by providing in one magnetic circuit a "star" and a "delta" path.
- If a symmetrical 3-phase transformer, in **Fig.10(d)**, is considered, it is seen that the limb flux Φ_l splits into two **yoke fluxes** Φ_y differing in phase each by 30° with respect to the limb flux, and requiring 5th and 7th harmonic currents displaced respectively by $\pm 5 \times 30^o = 150^o$ and $\pm 7 \times 30^o = 210^o$, with respect to the current components in the limb. The two yokes together by symmetry, take 5th and 7th harmonics excitation in **phase opposition** to that of the limb to which they are magnetically connected, and the combination may be devoid of harmonic magnetizing currents if the **reluctances are suitably chosen** unless high saturation densities are used, in which case harmonic yoke excitation can be provided by a yoke winding.
- This star-delta arrangement is inherent in the 5-limb core in **Fig.10(e),** and could also be obtained in a 3-limb core in **Fig.10(f)** by **yoke slitting**. A practical method would be to **wind round** the yokes a delta-connected winding to force the fluxes into the required relationship in spite of any difference in reluctance.
- In **Fig.10(f)**, there is in addition an auxiliary star-connected winding interlinked with the delta-winding on the yokes to aid production between the limbs and yokes of the 5th and 7th harmonic currents.

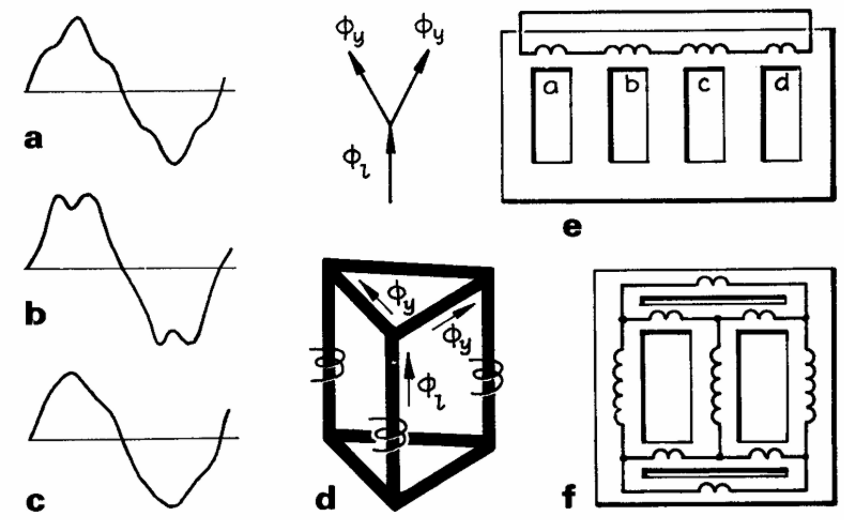


Figure 10: Harmonic Current Compensation

6. Transformer Inrush

Short duration switching currents and line-surge voltages may occur in transformer operation, of such magnitude as to impose high mechanical stress or insulation damage.

When an **unloaded transformer** is switched on, it acts as a **nonlinear inductor**. The behaviour depends on the magnitude, polarity and rate of change of the applied voltage at the instant of the switching.

Assume the core is initially unmagnetized and that the switch is closed at the **voltage peak**. The balancing EMF demands a flux having maximum rate of change, so that the flux starts to grow from zero in the required direction, as also does the magnetizing current. Such conditions are those of normal no-load operation, in **Fig. 11(a)**, and there is no transient.

If however, the switch closes at a voltage zero as in **Fig. 11(b)**, the voltage is positive during the first half-period, throughout which the flux must therefore increases, with a total change corresponding to that of a "normal" half-period. It therefore reaches a maximum of twice the normal peak, i.e. $2\Phi_m$, a phenomenon called the **doubling effect**. In subsequent half-periods the effect of losses rapidly reduces the flux waveform to symmetry about the time-axis.

Flux doubling implies a peak density of 2 x normal steady state value. With high saturation levels employed for modern transformers, the magnetizing current required for twice normal flux density is very great. For peak density of 1.4T, for example, the excitation is about 3kA-t/m; for twice this density it may be increased 100-fold. A normal magnetizing current of 0.5pu is for such condition raised to 5 pu, producing an electromagnetic force 25 pu of normal.

The doubling flux density will not be reached, for with such overcurrent the transformer resistance drop is significant, and further limitation is imposed by the **source impedance** of the supply network. The transformer switching transient is referred to **inrush current**.

Remanent flux in the core can aggravate the condition. If the primary voltage is applied at a zero instant and in such a direction that the rising field augments the existing remanent flux Φ_r as in **Fig. 11(c)**, the sinusoidal of flux variation has still the amplitude $2\Phi_m$, but is entirely offset from zero and rises to a peak value of $2\Phi_m + \Phi_r$.

Hysteresis loops in terms of flux and current are shown in **Fig.11(d)** for the cases in **Fig. 11(a)**, **(b)** and **(c)**. The first is normal and symmetrical, the second is uni-directional and the loop area is actually reduced. The third, for case **(c)**, shows **super-saturation** and an almost negligible hysteresis loss.

At **low densities** the flux is substantially confined to the core, but at **high saturation** the permeability of the core steel is so low that the core shares much of the flux with what are normally leakage paths. The peak magnetizing current approximates to that determined by the inductance L of the primary winding considered as an air-cored coil. The current peaks thereafter fall rapidly because the rate of decay is a function of the ratio R/L, where R is a large loss resistance and L is subnormal.

As the flux density reduces towards its normal excursions the loss resistance reduces but the inductance increases, slowing the rate of decay and causing current distortion to persist for several seconds, as shown in **Fig. 11(e)**.

A condition akin to remanent flux can occur in a 3-limb transformer if one primary terminal is energized later than the other two, and in the **neutral connection** a large **inrush current** characterized by an **exponentially decaying DC** may occur.

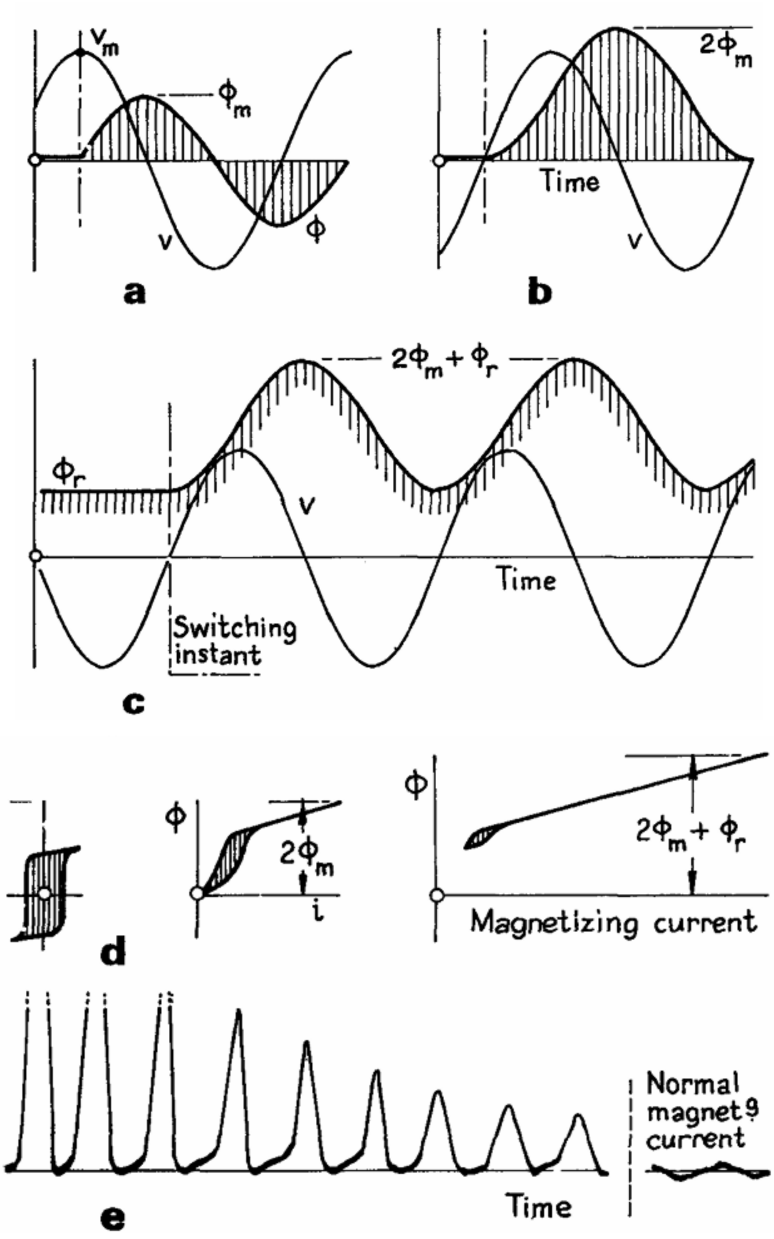


Figure 11: Inrush Current Transient

7. Three Phase Connection

With windings that can be connected in **star**, **delta**, **zigzag** or **vee**, and with primary, secondary, **tertiary** and **auto-connected** windings, the variety of possibilities is considerable.

A choice is possible between a 3-ph unit and a bank of three 1-ph transformers. The unit may cost about 15% less than the bank and will occupy much less space. There is little difference in reliability, but it is cheaper to carry spare 1-ph than 3-ph transformers if only one installation is considered.

The choice between **star** and **mesh** connection merits consideration in each case.

Star connection is cheaper, since **mesh** connection needs more phase turns and more insulation. The difference is small, however, at voltages below 11 kV. With very high voltages a saving of 10% may be effected, mainly on account of the insulation. An advantage of the star-connected winding with **earthed neutral** is that the **voltage from the core to the case** (frame or earth) is limited to 58% of the line voltage, whereas with a delta-connected winding the earthing of one line (due to fault) increases the maximum voltage between windings and core to the full line voltage. Technically the mesh-connected primary is essential where the LV secondary is a star-connected four-wire supply to mixed 3-ph and 1-ph loads.

Nomenclature

HV terminals are designated by capital letters, e.g. ABC, and LV terminals have the small letters, e.g. abc. Neutral terminals precede line terminals.

Each winding has two ends designated by subscripts 1 and 2. If there are intermediate tappings these are numbered in order of their separation from end 1. An HV winding with four tappings is designated A_1, A_2 ... A_6, A_1 and A_6 being the ends of the phase winding. If the e.m.f. on the HV phase winding has at a given instant the direction A_1A_2, then at the same instant the e.m.f. direction in the associated l.v winding is a_1a_2. The diagrams in **Fig. 12** show the connections and e.m.f. phasors for a number of common arrangements.

Polyphase transformers are all allotted symbols giving the **type of phase connection** and the ***angle of advance*** in passing from the phasor representing the EMF of the HV winding to that of the LV phasor at the corresponding terminal.

The angle is indicated in terms of a clock face, the HV phasor being at 12 o'clock (zero) and the corresponding LV phasor at the hour-hand number. Thus 'YzD3 11' represents a (HV star/LV zigzag/tertiary delta)-connected 3-ph transformer with the LV (secondary) EMF phasor in clockwise rotation at 11 o'clock, i.e. 30° in advance of the 12 o'clock position of the HV phasor. The groups into which 3-ph transformers are classified are as follows.

Group	*Phase displacement*	*Two-winding connections*		
1	Zero	Yy0	Dd0	Dz0
2	180°	Yy6	Dd6	Dz6
3	30° lag	Dy1	Yd1	Yz1
4	30° lead	Dy11	Yd11	Yz11

Connections -

Star/Star (Yy0 or Yy6)

- This is economical for small HV transformers as it **minimizes the turns/ph** and the **winding insulation**.
- With a star neutral available on both sides, it is possible to provide a **neutral connection**.
- **Triplen voltages are absent** from the line, and (unless there is a neutral connection) no triplen currents flow. The **neutral/voltage may oscillate**, and triplen voltages may be high in small-type units.

Delta/Delta (Dd0 or Dd6)

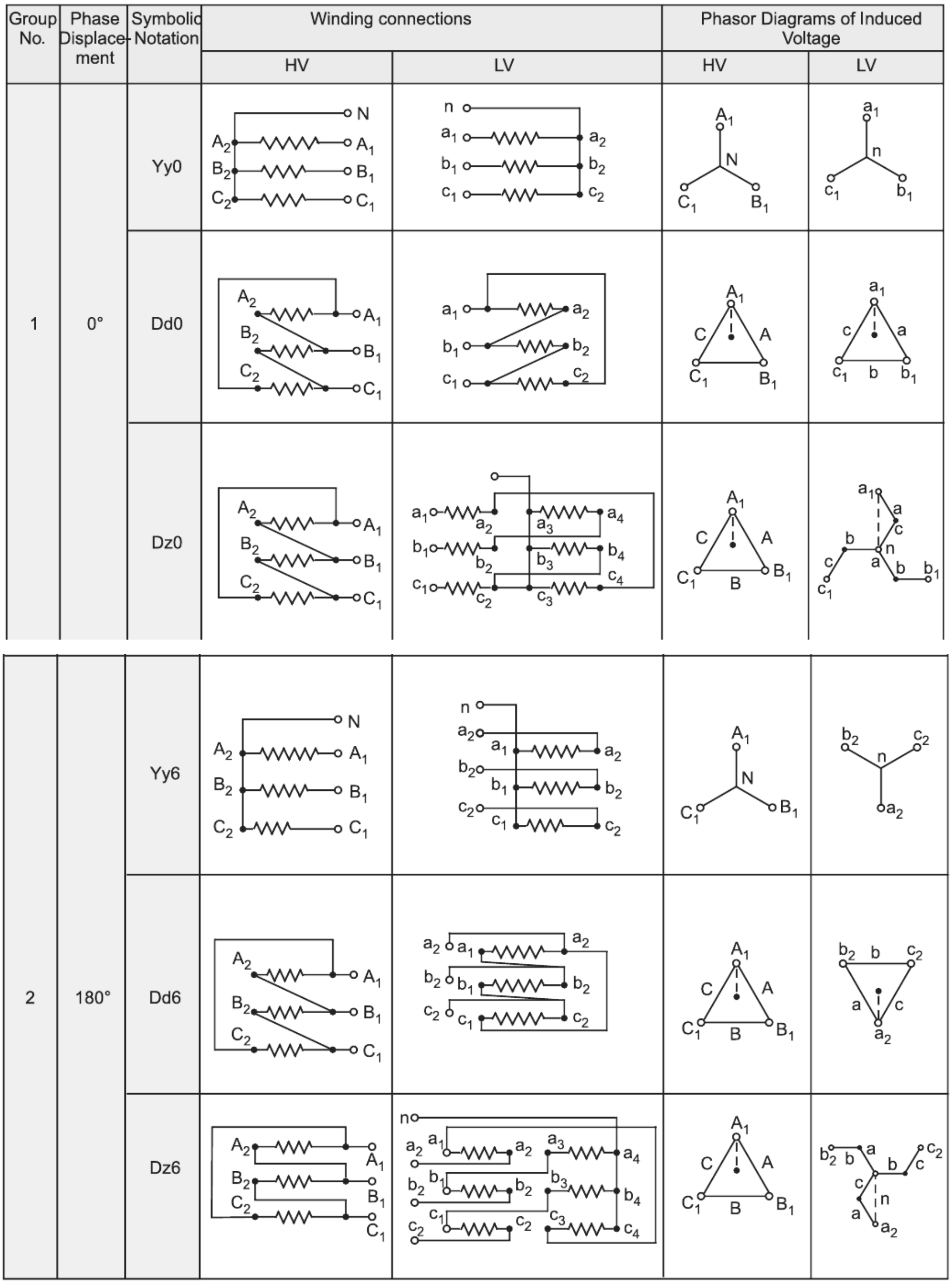
- These are suited to **large HV transformers** as it requires more turns/ph of smaller section.
- **Absence of a star point** may be a disadvantage.
- The **large load unbalance** can be tolerated, and triplen voltages are damped out.

Star/Delta (Yd or Yd)

- It has a strong point for a neutral to **serve mixed 1-ph and 3-ph loads**, and a **delta winding** that can **carry triplen currents** and so reduces the triplen voltage.

Interconnected-star (Zigzag or Yz11)

- The interconnected-star ('zigzag' star arrangement) **reduces triplen voltages** and is **not sensitive to unbalanced loading**.
- The star point is **earthed** to **comparatively low-voltage windings**, and as the phase voltages are composed of half-voltages with 60° displacement, 15% more turns are required for a given phase terminal voltage compared with a normal star.
- The connection is sometimes employed in **rectifier supply**.



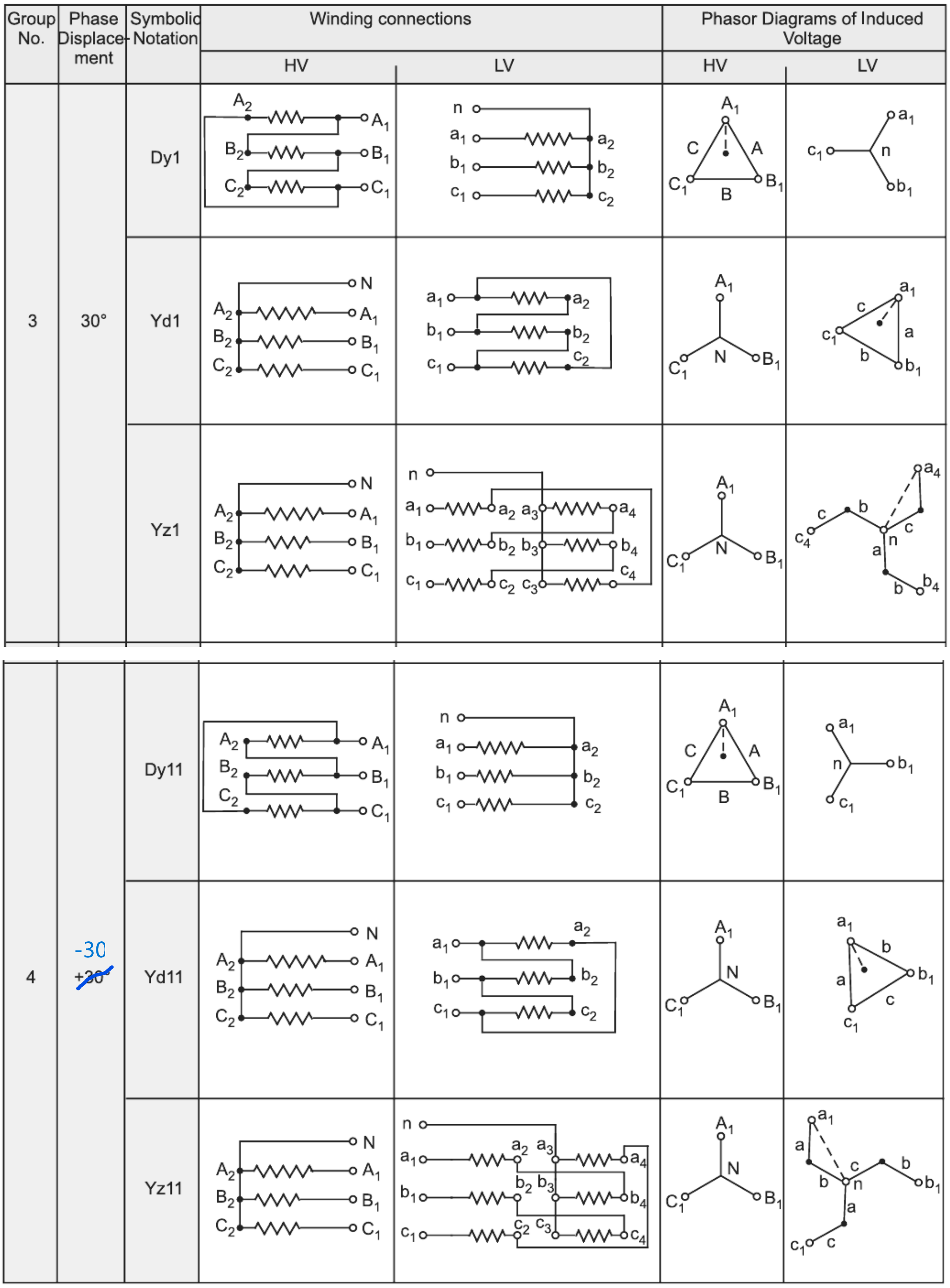


Figure 12: Standard Connection for 3-phase Transformer

8. Three Phase to Two Phase Connection

Tee-Tee Connection or Scott Transformer

This requires two transformers with different tappings and ratings, although they may be constructed for interchangeability.

- Consider **Fig. 13**, The N1 turn primary of a 1-ph ('main') transformer connected between terminals B and C of a 3-ϕ supply of line voltage V.

- The voltage across line A and the mid point S of BC is $\sqrt{3}/2\ V = 0.87V$. If the primary of the **Teaser transformer** with 0.87 N1 turns is connected across AS, the voltage per turn is the same for both main and teaser.
- Then two identical secondaries with N2 turns have terminal voltage of equal magnitude but phase quadrature.
- The 3-ph neutral point N is located on the teaser primary at one-third of the winding distance from S, where the voltage is V/$\sqrt{3}$= 0.58 V in AN and phase 0.29V in SN.
- Let equal currents I_2 at unity p.f. be taken from the two secondary phases, **Fig.13(a)**, and let magnetizing current be neglected. There must, in each transformer, be an **m.m.f. balance (a.k.a. ampere turn balance ATB)**. In the main transformer the primary balancing current is

$$\frac{2I_2N_2}{\sqrt{3}N_1} = 1.15I_2\left(\frac{N_2}{I_1}\right)$$

- The teaser primary current is in phase with the star voltage AN. The total current in the main primary is the resultant of two components: The first is $I_2(N_2/N_1)$ to balance the secondary current, the second is one half of the teaser primary current in either direction from S and C are shown in **Fig.13(a)**, taking N1 = N2 for clarity; their phasors, combining the rectangular components I_2 and $0.58I_2$, are co-phasal respectively with the star voltages NB and NC, and are equal to the teaser primary current. Thus for a balanced N-phase load of unity p.f. the 3-ph side is balanced.
- Case **(b)** is for a balanced 2-ph load of PF = 0.71. Again, the 3-ph load is balanced, and the main transformer loading is 15% greater than that of the teaser. Case **(c)** is for the unbalanced 2-ph load.

Le Blanc Connection

The Le Blanc connection is used when it is desired to connect a 3-ph winding to a 2-ph system.

- The 3-ph side consists of three windings in star or delta wound on an ordinary 3-limbed core. Suppose the 2-ph side is to have a phase voltage V. Then the limb ABC in **Fig. 14** are respectively with winding a1 of voltage 2/3 V; b1 of voltage 1/3 V with b2 of voltage V/$\sqrt{3}$, and c1 of voltage 1/3V and c2 of voltage V/$\sqrt{3}$. These are grouped to form two phases:
    - Phase I of 2 ph supply: a1, b1, and c1 in series; Phase II: b2 and c2 in series.
- The phasor sums show that the 2-ph side has equal voltages in phase quadrature. The purpose of b1 and c1 in phase I is to **balance the m.m.f.s around the three interlinked magnetic circuits** and to give 3-ph input balance for a balanced 2-ph load. The connection uses a standard 3-ph core.

Open Delta or Vee-Vee Connection

- The delta-delta connections of three transformers for a three-phase system are shown in **Fig. 15(a)**. If a fault develops at one of the transformers and the transformer was removed, and the consumer wants to continue three-phase supply as shown in **Fig. 15(b)**, then three equal three-phase voltages will be available at the secondary terminals at no-load. This method of **transforming three-phase power** by means of only **two one-phase transformers** is called **open delta** or **Vee-Vee connections**.
- The basis of operation of open-delta connections is because of the fact that the vector sum of any two of the line voltages in a balanced three-phase system is equal to the third line voltage.

Thus even though one of the transformer has been removed, the voltage between the terminals of the secondary to which load has been connected remains unchanged.

Advantages of Open-Delta or Vee-Vee Connections

- **Continuity of Supply**: If one transformer in a delta-delta bank fails, the remaining two can continue to supply three-phase power without interruption.
- **Economical**: For light loads, only two transformers may be installed initially. A third can be added later when the load increases.
- **Reduced Initial Cost**: Since only two transformers are used, the initial investment is lower.
- **Flexibility:** Open-delta systems allow easy expansion and maintenance without complete shutdown.

Output Considerations -

- Consider the **effective output** from open-delta systems.

  Power in delta systems = $\sqrt{3}V_L I_L \cos\phi = \sqrt{3}V_L\sqrt{3}I_{ph}\cos\phi = 3V_L I_{ph}\cos\phi$

  Power in Vee vee connection = $\sqrt{3}V_L I_{ph}\cos\phi$

  Hence,

$$\frac{\text{Power in Vee vee connection}}{\text{Power in delta systems}} = \frac{\sqrt{3}V_L I_{ph}\cos\phi}{3V_L I_{ph}\cos\phi} = 57.7\%$$

- It is assumed that the power factor in both systems is the same, but in practice the power factor in V-V connects is less than the actual power factor of the load.
- The two transformers constituted 66.7% of the installed capacity of the three, but they are only able to deliver 57.7% of the three in open delta connection. The reduced output is due to the reduced power factor in Vee-Vee connection.

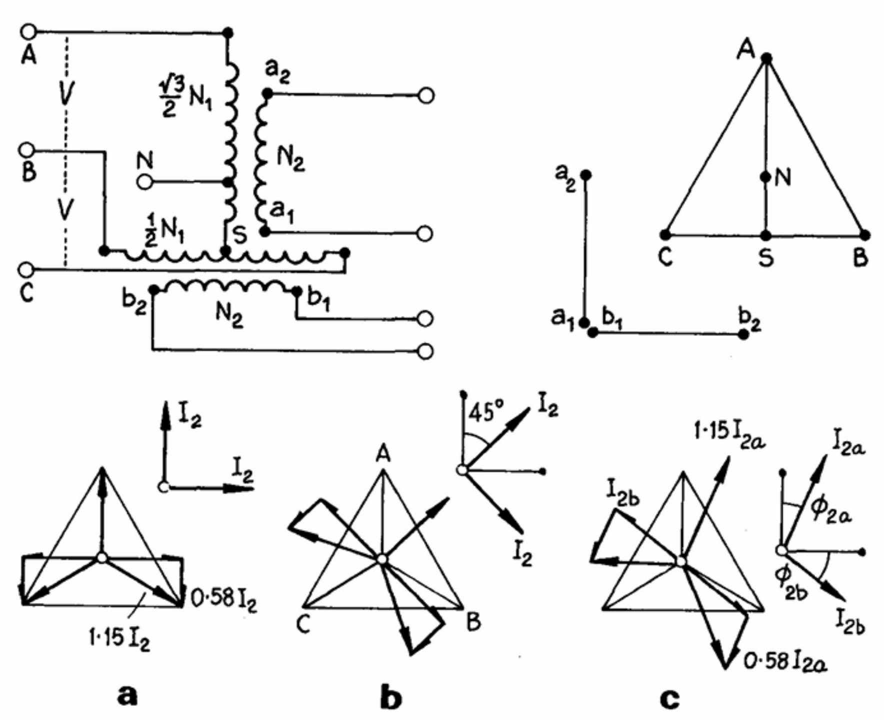


Figure 13: Scott Transformer

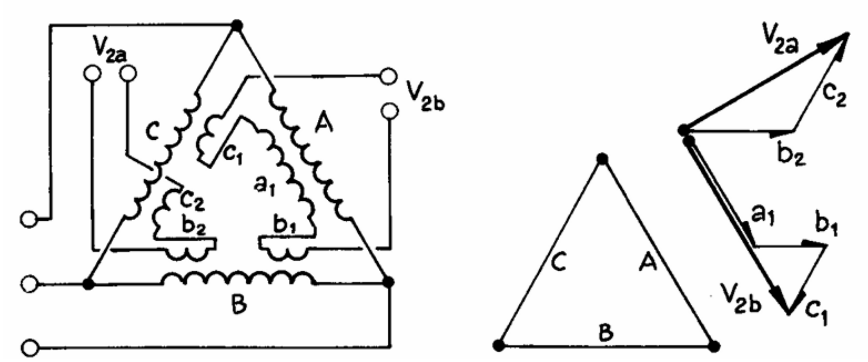


Figure 14: Le Blanc Transformer

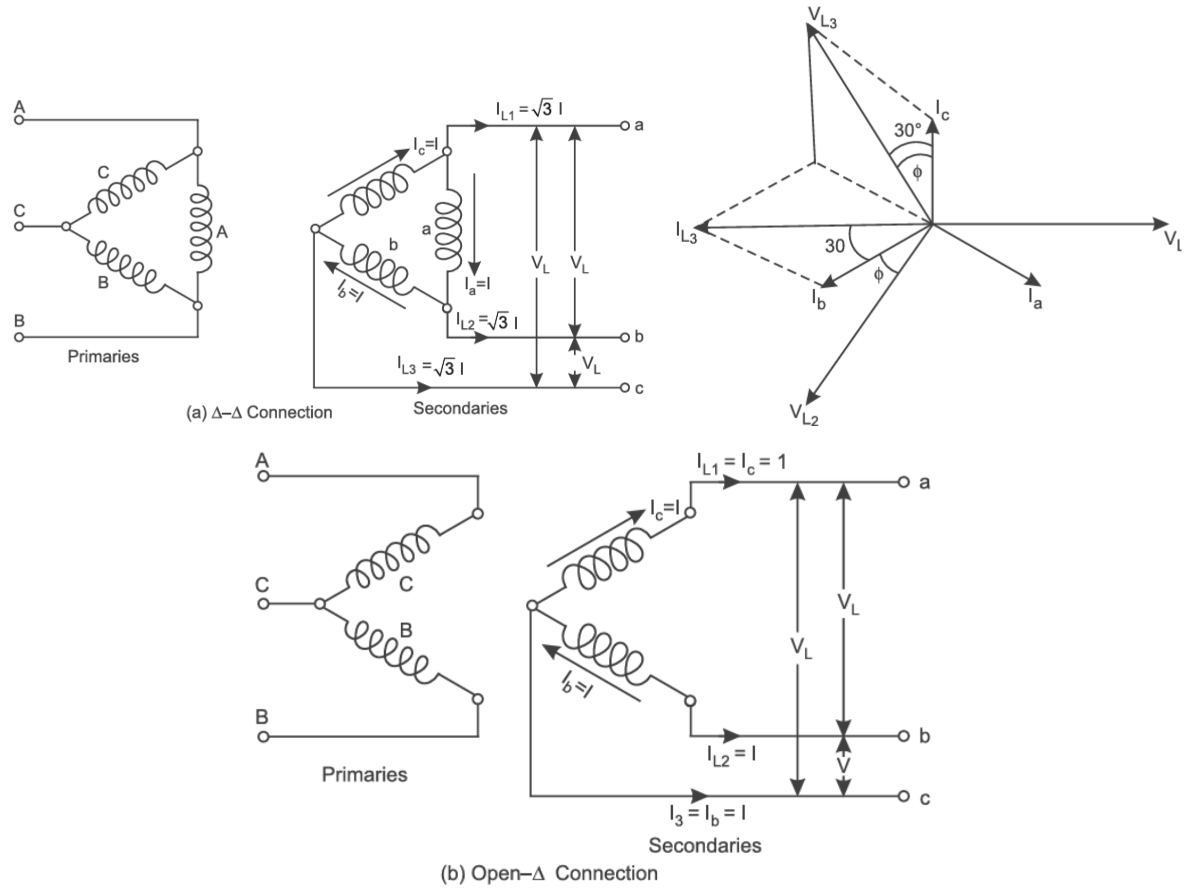


Figure 15: Open Delta or Vee-Vee Connection

Single Phase Connection -

Single-phase power pulsates twice per period, whereas a balanced 3-ph load has constant power over the three phases. It is therefore not possible to obtain a balanced 3-ph load from a single-phase source.

Fig.15 shows simple direct methods. The actual division of the single-phase current can be adjusted to any proportion between 1:1:0 and 1:1/2:1/2.

- (a) is the simplest connection, one phase of a Scott connection.
- (b) is one phase of a Scott connection.
- (c) is an open delta.
- (d) uses both units of the two-phase Scott arrangement.

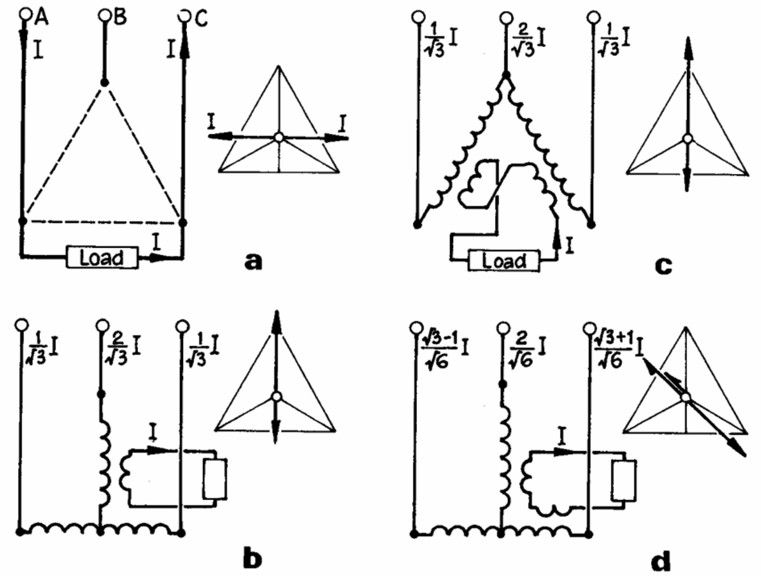


Figure 15: Single Phase Connection

9. Auto Transformers

- Auto transformers have windings common to both primary and secondary sides, so that the input and output circuits are **electrically connected** as one continuous winding. This is useful where the voltage ratio is not less than 2:1, where **electrical isolation** of primary circuit to secondary circuit is not essential, with application as **starters** and **interconnection** of industrial transmission lines. The advantages are smaller size, reduced rating and lower losses.
- Neglecting magnetizing current, let the 1-ph auto-transformer in **Fig. 16** have N1 primary turns tapped at N2 for a lower-voltage secondary. For an output of I_2 at voltage V_2, the input is $I_1 = I_2(N_2/N_1)$ at voltage $V_1 = V_2(N_1/N_2)$ if the losses can be ignored. The current in the common part of the winding is $I_2 - I_1$ so that here the conductor section may be smaller. With the same conductor current density throughout, the ratio G_a/G_2 of conductor material in the auto and two-winding transformers is

$$\frac{G_a}{G_2} = \frac{I_1(N_1 - N_2) + (I_2 - I_1)N_2}{I_1N_1 + I_2N_2} = 1 - \frac{V_2}{V_1}$$

The saving in materials and cost is less than this, as the core is nearly as large as in the two-winding transformer. For a voltage ratio of 2:1, about 50% **saving in winding materials** may be obtained, and the overall cost of the unit may be 60%-70% of that of a normal transformer of the same input rating.

Equivalent Circuit

Fig. 16 shows at **(a)** the equivalent circuit ignoring magnetizing and core-loss current, and at **(b)** the corresponding phasor diagram. For a load phase angle ϕ the primary and secondary voltages are approximately:

$$V_1 = E_1 + I_1(R_1\cos\,\varphi + X_1\sin\,\varphi) + (I_1 - I_2)(r_2\cos\,\varphi + x_2\sin\,\varphi)$$

$$V_2 = E_2 - I_2(r_2\cos\,\varphi + x_2\sin\,\varphi)$$

Combining these and writing $E_2 = \frac{N_2}{N_1}E_1$ and $I_2 = \frac{N_1}{N_2}I_1$ yields the relation:

$$V_1 = \frac{N_1}{N_2}V_2 + I_1(r_1 + r_2')\,\cos\varphi + I_1(x_1 + x_2')\,\sin\varphi$$

where:

$$r_2' = \left(\frac{N_1}{N_2} - 1\right)^2 r_2 \text{ and } x_2' = \left(\frac{N_1}{N_2} - 1\right)^2 x_2$$

And R_1 and X_1 are the equivalent resistance and leakage reactance referred to the primary. A secondary short circuit test gives:

$$Z_1 = R_1 + jX_1$$

as the ratio of the applied primary voltage and input current, just as the two-winding transformer. The magnetizing branch can be obtained with an open circuit test.

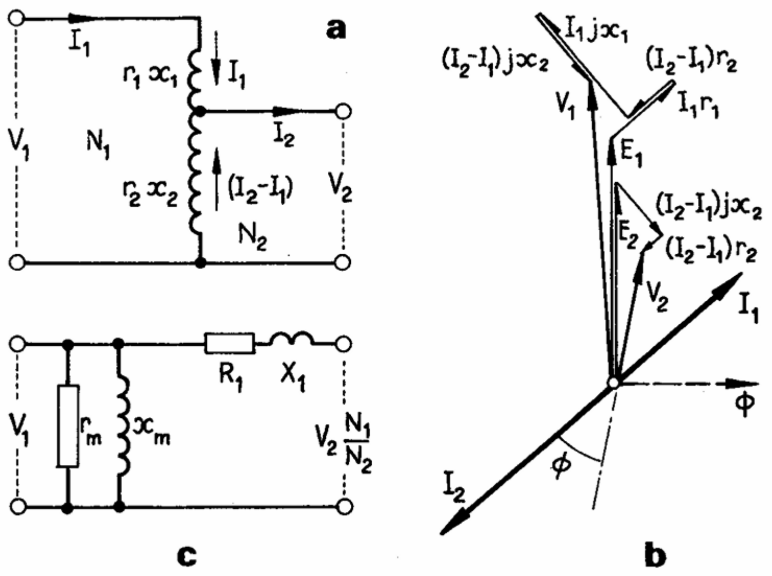


Figure 16: Auto-Transformer

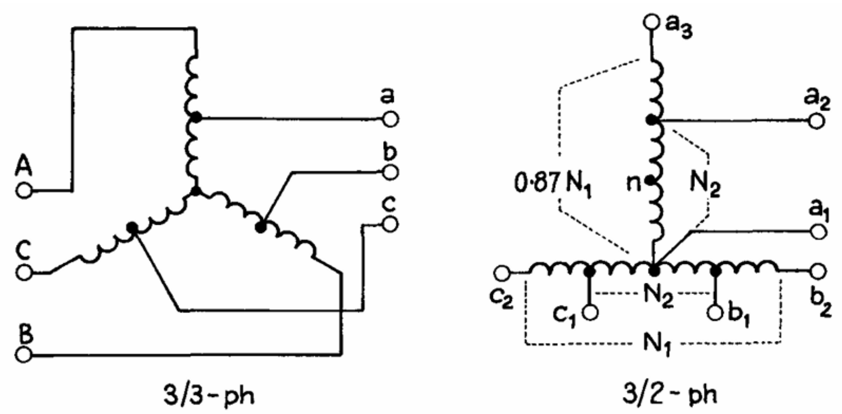


Figure 17: Three-phase Auto-Transformer

10. Three-winding connection

Transformers may be built with tertiary windings (i.e. additional to the primary and secondary) for one of the following reasons:

- To supply a small additional load at a different voltage. The tertiary winding is called **auxiliary winding**.
- To supply phase-compensating devices (e.g. capacitors/ reactors) requiring a different voltage or connection.

- In star/star or star/zigzag transformers, a delta-connected tertiary winding reduces the zero-phase-sequence impedance and allows adequate earth-fault current to flow for the operation of protective gear, and it limits voltage unbalance when the main load is asymmetric. The delta-winding is termed a **stabilizing winding**.
- To indicate voltage in a HV testing transformer.
- To load-split windings (generators).

Tertiary windings are frequently **delta-connected**: consequently, when faults and short-circuits occur on the primary or secondary sides (particularly between lines and earth**), considerable unbalance of phase voltage may be induced**, compensated by **large tertiary circulating currents**. The reactance of the winding must be such as to **limit the circulating current** to that which can be carried without overheating.

With given secondary and primary load currents I2 and I1 and with windings of N1, N2 and N3 turns/ph, the tertiary current I3 is readily found by **phasor summation** of the secondary and primary magnetizing m.m.f.s or equivalent currents (by ampere turn balance).

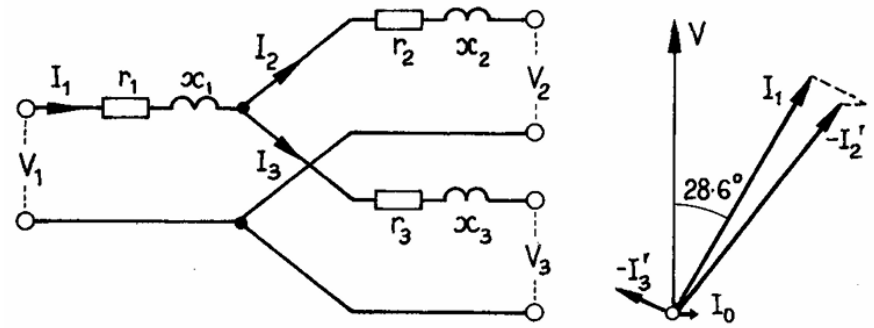


Figure 18: Three Winding Transformer

Equivalent Circuit

Properly interpreted, a three-winding transformer can be represented by the equivalently 1-ph circuit of **Fig. 18**, each winding assumed to have an **equal resistance** and **separate leakage reactance**. All the values are reduced to a *common-voltage* and *common-rating* basis, and the primary input, the secondary output is not in general substantially the same as the primary secondary component voltages used in the two-winding transformer are no longer valid: the total power between secondary and tertiary is, in fact, arbitrary.

Consider the primary and secondary operating as a two-winding transformer. The total **per-unit resistance voltage** $Ir_{12}$is the sum of the drops in the two windings separately. A similar statement is true for any pair of windings, and can be stated also in appropriate terms for the reactance drop. Thus we have

$$Ir_{12} = Ir_1 + Ir_2, \qquad Ix_{12} = Ix_1 + Ix_2,$$
$$Ir_{23} = Ir_2 + Ir_3, \qquad Ix_{23} = Ix_2 + Ix_3,$$
$$Ir_{13} = Ir_1 + Ir_3, \qquad Ix_{13} = Ix_1 + Ix_3,$$

for the possible combinations of pairs of circuits taken together as two-circuit transformers. Using these relations,

$$r_1 = \frac{1}{2}(r_{12} + r_{13} - r_{23}) = \frac{1}{2}\Sigma r - r_{23},$$

The other corresponding relations are similar, whence

$$r_1 = \frac{1}{2}\sum r - r_{23} \quad x_1 = \frac{1}{2}\sum x - x_{23},$$
$$r_2 = \frac{1}{2}\sum r - r_{13} \quad x_2 = \frac{1}{2}\sum x - x_{13},$$
$$r_3 = \frac{1}{2}\sum r - r_{12} \quad x_3 = \frac{1}{2}\sum x - x_{12},$$

From a series of normal two-winding short-circuit tests, these expressions give, for the assigned voltage and rating, the ohmic values r_1, r_2, x_1,and x_2, as shown in **Fig. 18** taking completely into account all mutual effects.

Suppose the secondary to be short-circuited, the **tertiary being left on open circuit**. The total voltage applied to the primary will be the sum of the primary and secondary short-circuit voltages. A voltmeter across the tertiary output terminals is connected in effect across the junction of primary and secondary: it therefore indicates the **secondary voltage** (after adjustment to the common voltage-base). In general, the voltage across one circuit when a second supplies short-circuit apparent power to the third, is the combined voltage of the short-circuited winding.

Regulation

This is best considered from the individual contribution of each winding, the total for two windings in combination being the algebraic (not phasor) sum. For the primary and secondary,

$$e_1 = k(\epsilon_{r1} \cos \phi_1 + \epsilon_{x1} \sin \phi_1)$$

Summation of the **part-regulations** of two circuits together is an **addition** if power flows from one to the other, a **subtraction** if not. For a transformer with primary fed and both **secondary and tertiary loaded**, the regulations become:

$$\epsilon_{12} = \epsilon_1 + \epsilon_2; \;\; \epsilon_{13} = \epsilon_1 + \epsilon_3; \;\; \epsilon_{23} = \epsilon_2 - \epsilon_3, \epsilon_{32} = \epsilon_3 - \epsilon_2$$

11. Zero Sequence Impedance

The impedance seen by positive sequence and negative sequence voltages is the same, as the phase sequence order makes no difference to the connection. When zero sequence voltages or currents are present, account must be taken of winding connections, earthing, and, in some cases, the constructional form.

Fig. 19 sets out the connections and zero-sequence conditions in **A** and for two- and in **B** for three-winding transformers, respectively. In A, the reactance $x_{12} = x_1 + x_2'$; In B, the reactance x_{12}, x_{23}, x_{13} has the same significance in Section 10.

Two-winding Connections

- **Yy (both star points isolated)** - There is no earth circuit. An infinite impedance is offered to zero sequence currents.
- **Yy (secondary neutral earthed)** - An earth return path is still not found in zero sequence if the transformer has **no phase magnetic interlinkage** (shell type or screen 1-phase units), but in a **core-type** either or both the primary and secondary side if the transformer has magnetic flux that interlinking gives complete circuit for zero sequence currents, a **high-reactance path** outside the core.
- **Yy (both neutrals earthed)** - Provided that the primary circuit is complete for zero sequence currents, a secondary neutral that will give rise to zero sequence components when both neutrals are earthed in the primary circuit; and similarly for an **earth return path** in the secondary. In core-type, zero sequence currents is three times the reactance of the 3-phase windings in

parallel, i.e. the zero sequence reactance value is 1/3 of the identical to the positive sequence impedance values. With 3-ph core-type transformers the effect described may be used to determine the zero-sequence reactance value.

- **Yd -** For a primary earth fault, zero sequence currents can flow to from the primary neutral (if earthed), but they can circulate within the delta.
- **Dy (neutral earthed)** - For a secondary earth fault the zero-sequence reactance is infinite. For a primary earth fault, comparing currents flow in the secondary delta. The zero-sequence reactance is 1/3 of the positive sequence impedance.
- **Dd -** No zero sequence current, but to form either side or in the delta.
- **Zigzag transformer –** To positive sequence and negative sequence reactance, the reactance is infinite; To zero sequence current, the reactance is due only to leakage as the two MMF on each limb are self-cancelling.

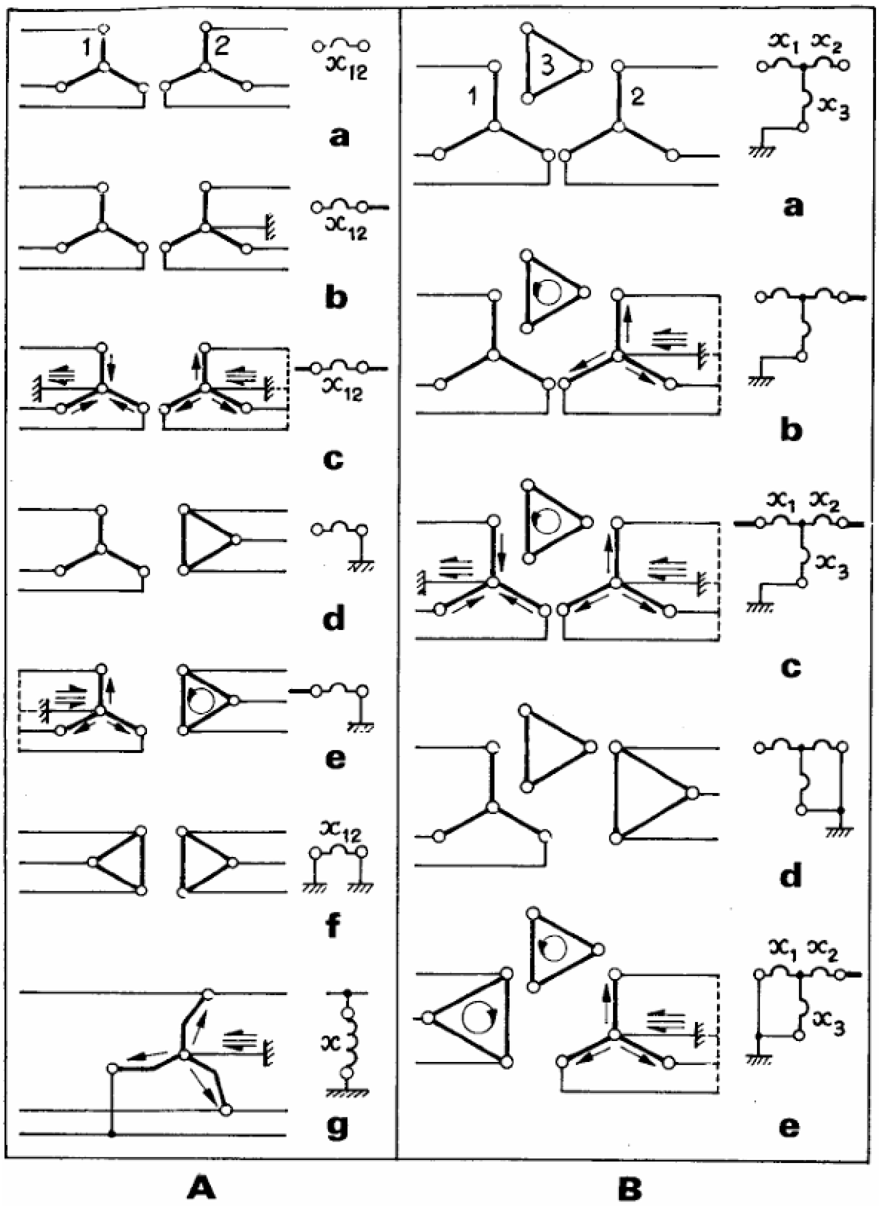


Figure 19: Zero Sequence Connection

12. Tap Changing

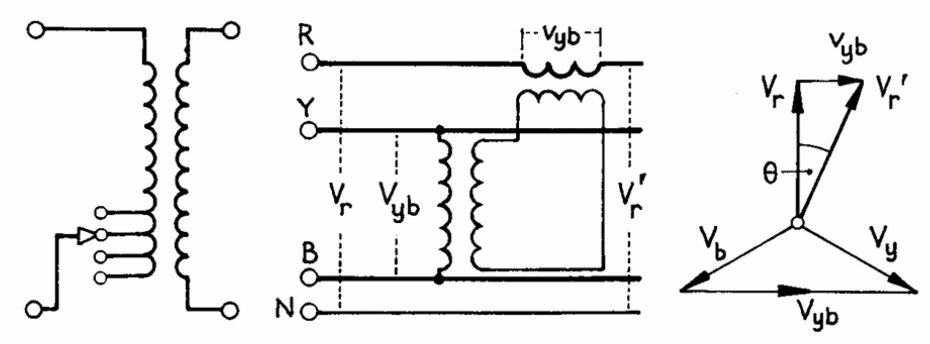


Figure 20: Voltage Control by Tap Changing

Simple up or down control voltage regulation by tapping is shown in (a), while (b) is for a control of phase angle in which a voltage v_{yb} derived from the line voltage V_{yb} gives an output voltage $V_r{}'$ differing from V_r by the angle θ. It can be used for control of reactive power flow on interconnected feeders.

The **principal tapping** is that to which the rating of the winding is related. A **positive tapping** increases torque, and a **negative tapping** less, turns than those of the principal tap.

Tap-changing may be arranged in accordance with one of three conditions, namely:
(i) voltage variation with constant flux and constant turn-emf i.e.:
(ii) with varying flux; and
(iii) a mixture of (i) and (ii).
In (i) the percentage tapping range is identical with that of voltage variation.

Location of the tapped part of the winding is partly a constructional question:

- With tappings near the neutral ends the number of bushing insulators is reduced.
- With tappings near the line ends the insulation conditions between phases are eased.
- Where a large voltage variation is required, tappings should be near the center of the phase windings to reduce magnetic asymmetry, but this arrangement cannot be used on LV windings placed next to the core.

It is not possible to tap other than an **integral number of turns**, and this may cause difficulty with LV windings. A 260 V phase winding with 15 V/turn cannot be tapped closer than 5%. It is consequently necessary to tap the HV windings, a further advantage in a step-down transformer being that, on light load with lowest secondary voltage, the maximum number of turns is included on the HV side, reducing the e.m.f. per turn, the flux density and the core loss.

On-load Tap-changing

The essential feature is the maintenance of circuit continuity throughout the tap-change operation. **Momentary connection** must be made simultaneously to two adjacent taps during the transition, and the **short-circuit current** between them must be limited by some form of impedance. Inductors have been used for this purpose, but in modern designs the **current limiting** is almost invariably obtained by means of a **pair of resistors**. **Fig. 21** shows the method, which has a straightforward switching sequence. **Back-up main contactors** are provided which short-circuit the resistors for normal operation.

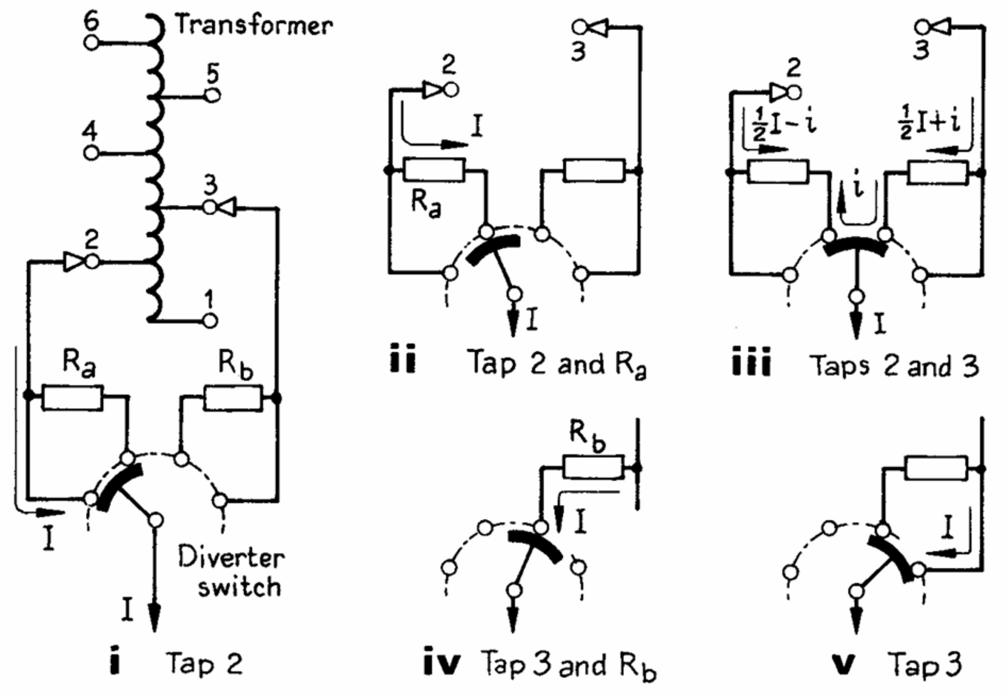


Figure 21: On Load Tap Changing

One winding tap is required for each operating position. The transition time must be kept as short as possible to avoid excessive heating of the transition resistors. To ensure continuous switching even with failure on auxiliary control supply, **energy storage mechanisms** are used. These are usually spring-operated, to reduce the time that the resistor is in circuit to a few periods, and are charged by the motor drive. The current flowing through the resistors during the transition is limited to a few hundred amperes. Such tap-changer is compact, and the high speed break limits the contact wear.

Two switching arrangements are used for on-load tap-changers:

1. A **separate contactor** for each winding tap. The contactors are operated by a **camshaft** to ensure the correct sequence.
2. The winding taps are connected to a **selector switch** with an associated pair of contacts. The **make and break** takes place at the selector switch contacts and some degree of carbonization and wear is unavoidable. Such arrangement is common for transformer up to 20MVA.
3. For very large transformers the tap selector switches do not move when carrying current: make and break is carried out by **two separate diverter switches**, usually in a separate compartment to **minimize main-oil pollution by carbon**. The diverters are mechanically interlocked with the selectors, which move only (when not current loaded) to connect with the sequence of tap connections. **Vacuum switches** may be used as diverters, and as they do not contaminate oil they can be mounted in the same compartment as the selectors.

On-load tap-changer control gear can vary from simple **push-button initiation** to complex **automatic control** of as many as four transformers operating in parallel together. The object of automatic voltage regulation (AVR) control is to **maintain a set voltage level** within a specified tolerance, or to raise it to within a set tolerance after a power cut.

The main components are an automatic voltage regulator, a time-delay relay, and compounding elements. The time-delay prevents initiation of a tap-change by a **minor transient voltage fluctuation**: it can be set for a delay up to 60 s.

Line Drop Compensation - This is essentially an artificial transmission line consisting of adjustable resistor and inductor elements combined with control and measurement elements to provide a signal to the voltage relay coil. Variation of the load current then gives compensation for the line drop through the **on-load tap-changing** equipment.

13. Parallel operation

The satisfactory performance of transformers connected on both sides in parallel requires that they essentially feature the **same polarity**, the **same phase-sequence**, and **zero relative phase-displacement angle**; a **near identity of voltage ratio**; and a **limited disparity in per-unit impedance**.

Polarity - This can either be short-displacement. If wrong, it results in a dead short circuit.

Phase-sequence and terminal phase-displacement –

These are seen with polyphase transformers. Only a few of the possible connections can be worked in parallel without excessive circulating current on small loads; for example, the secondary voltages of star/star and star/delta transformers may have a phase difference of 30°, making parallel connection inadmissible. The various connections produce various phase displacements and phase sequences: the phase displacements produced by tappings, but phase divergence cannot be compensated.

The phase-sequence must be identical for two parallel transformers. If three secondary terminals $a_1b_1c_1$ of transformer 1 are to be paralleled with three secondary terminals $a_2b_2c_2$ of transformer 2, the polarity and ratio being correct, then a_1 will may be connected to a_2. If the result is a potential difference across b_1b_2 or c_1c_2, then the phase-sequence differs.

The following are typical of the connections for which, from the viewpoint of sequence and phase displacement, transformers can be connected in parallel:

Transformer 1:	Yy	Yd	Yd
Transformer 2:	Dd	Dy	Yz

Thus all transformers in the same group can be paralleled.

Further, transformers with a +30° and a -30° angle can be paralleled by reversing the primary and secondary phase-sequence of one.

Voltage Ratio -

Voltage ratio (not necessarily precisely the same as equal turns-ratio) is necessary to avoid no-load circulating current when transformers are in parallel on both primary and secondary sides. The leakage impedance being low, a small voltage difference may cause considerable no-load circulating current and additional I^2R loss. On load, the circulation is masked but may cause overcurrent on one transformer when the parallel group is loaded to the combined output rating.

Leakage Impedance -

The leakage impedances of transformers required to operate in parallel may differ in magnitude and in quality (i.e. in reactance/resistance ratio). It is necessary also to distinguish between per-unit and ohmic impedance: the currents carried by two transformers are proportional to their ratings if their ohmic impedances are inversely proportional, and the per-unit impedances are directly proportional to those ratings.

A difference in the quality of the per-unit impedance results in a divergence of phase angle of the two currents, so that one transformer will be working with a higher, and the other with a lower, power factor than that of the combined output.

Load-sharing

Equal Voltage Ratios

If the voltage ratios are equal and the voltages coincident in phase, both sides of a pair of transformers can be connected in parallel and no current will circulate between them on no load. Neglecting magnetizing admittance, the case can be represented by the equivalent circuit of **Fig. 22(a)**, in which the leakage impedances (with respect to either primary or secondary side) are parallel connected.

Let the corresponding impedance be Z_1 and Z_2, and let the two currents be I_1 and I_2 (at a common voltage V) which together provide the total load current I. The volt-drop is

$$v = I_1 Z_1 = I_2 Z_2 = I Z_{12}, \qquad Z_{12} = Z_1 \parallel Z_2 = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

The total load current is

$$I = I_1 + I_2$$

Hence

$$I_1 = I \frac{Z_{12}}{Z_1} = \frac{Z_2 I}{Z_1 + Z_2} \quad \text{and} \quad I_2 = \frac{Z_1 I}{Z_1 + Z_2}$$

Multiplication by V gives the transformer loadings $S_1 = VI_1$ and $S_2 = VI_2$, and the combined load $S = VI$. Then

$$S_1 = S \frac{Z_2}{Z_1 + Z_2}, \text{and} S_2 = S \frac{Z_1}{Z_1 + Z_2}$$

The load division holds for per-unit loads and leakage impedances provided that all are expressed with reference to a common base.

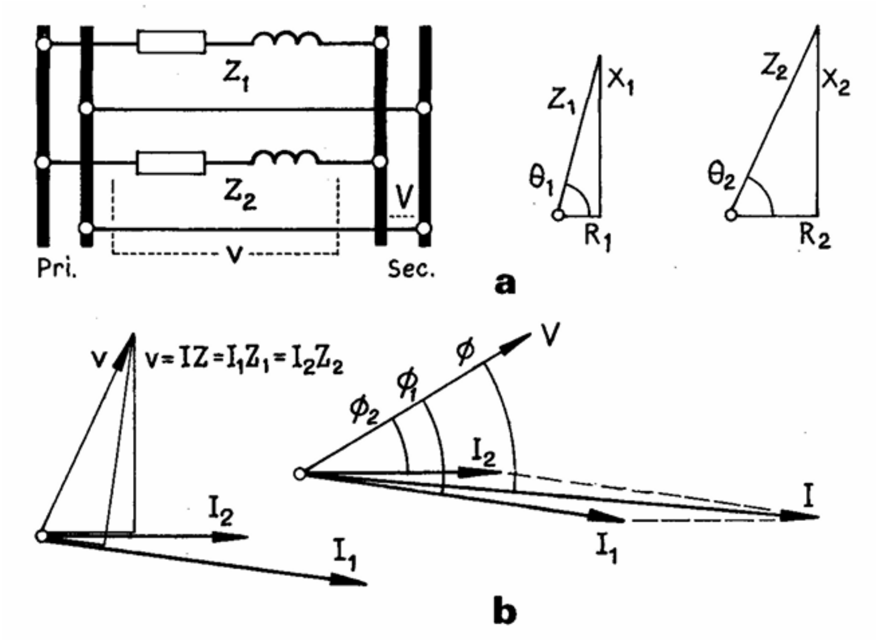


Figure 22: Parallel Operation

Fig. 22(a) shows the leakage impedance $Z_1 = R_1 + jX_1$ with the angle $\theta = \tan^{-1} X_1/R_1$ and $\mathbf{Z_2}$ in similar terms. The currents (b) must be such as to make I_1Z_1 equal in magnitude and phase to I_2Z_2 and the sum to the total load current $I = I_1 + I_2$. The actual output phase angles $\boldsymbol{\phi_1}$ and $\boldsymbol{\phi_2}$ differ $(\theta_1 - \theta_2)$.

Unequal Voltage Ratios -

With the notation in the previous discussion, The no-load secondary e.m.f.s, then

$$v = IZ_{12} = (I_1 + I_2)Z_{12}$$

as before. The transformer e.m.f.s are equal to the total voltages in their respective circuits, i.e.

$$E_1 = I_1Z_1 + IZ, E_2 = I_2Z_2 + IZ$$

whence

$$E_1 - E_2 = I_1Z_1 - I_2Z_2$$

Now no load with **I = 0** and given units to give the two transformers the circulating current

$$I_1 = -I_2 = \frac{E_1 - E_2}{Z_1 + Z_2}$$

and by substitution the two currents become

$$I_1 = \frac{E_1Z_2 + (E_1 - E_2)Z}{Z_1Z_2 + (Z_1 - Z_2)Z}, \qquad I_2 = \frac{E_2Z_1 - (E_1 - E_2)Z}{Z_1Z_2 + (Z_1 - Z_2)Z}$$

Normally E_1 and E_2 are co-phasal (or very nearly so); but current division equation could be used to show the severe results of incorrect parallel connection of 3-ph transformers with a phase-angle difference.

The loading-sharing of the number (or parallel arrangement) can more readily be found by applying the **Millman Theorem** (or **Parallel Generator Theorem**).

Given a common terminal voltage V of a number of transformers or generators, connected in parallel across a load impedance Z, and having respective internal impedances Z_1, Z_2, Z_3 ... is $V = I_{SC}Z_0$ where I_{SC} is the sum of the generator short-circuit currents and Z is the parallel sum of $Z, Z_1, Z_2, Z_3, \ldots$

$$V\left(\frac{1}{Z} + \frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3} + \cdots\right) = \frac{E_1}{Z_1} + \frac{E_2}{Z_2} + \frac{E_3}{Z_3} + \cdots$$

The square bracket is 1/Z and the right-hand side is the sum of the individual short-circuit currents.

Reactive Power Absorption -

Operating parallel transformers on different ('staggered') tappings is a way of **absorbing lagging reactive** power at points in a supply network.

In the case of two transformers each of leakage impedance $Z = R + jX$, with tappings staggered to give a voltage difference ΔV, the circulating apparent power is

$$S_c = \frac{\Delta V^2}{2Z}$$

For example, two 125 MW units each with R = 0.004 and X = 0.125 p.u., operate on load with 5% voltage difference. (2 1/2% up and down), circulate 0.05^2 / 0.25 = 0.01 p.u apparent power, i.e. 1.25MVAr. The additional I^2R loss is 0.04MW or 8% of full load I^2R.

14. Special Applications

1. Rectifier Transformers

Distortion of the current waveforms in transformers feeding power rectifiers increases the r.m.s. value for a given mean current and also the transformer apparent-power requirement and that for a sinusoidal load current. Further, large rectifier transformers produce some voltage waveform distortion on supply systems, and for installations of 1000 kV upward it may be desirable to limit distortion by increasing the number of pulses. The seven connections for 3-phase equipment in **Fig. 23** are classified:

1. **In accordance with number of pulses** (i.e. commutations per unit time)
2. **As single- or double-way** (i.e. transformer secondary currents either in one direction only or bidirectionally)
3. **Whether or not a bridge connection or an interphase is used –**
   - (a) **Three-pulse: single-way;** delta or star primary, zigzag secondary
   - (b) **Six-pulse: single way;** delta or star primary, double star secondary; interphase inductor
   - (c) **Six-pulse: single-way;** star primary with isolated star-point, diametral secondary
   - (d) **Six-pulse: double-way;** delta or star primary with isolated star-point, bridge secondary
   - (e) **Twelve-pulse: single-way;** delta or star primary, quadrature zigzag star secondary; three (or two) interphase inductors
   - (f) **Twelve-pulse: double-way;** delta or star primary, delta or star secondary; two parallel 3-ph bridges, interphase inductor
   - (g) **Twelve-pulse: double-way;** delta or star primary, delta or star secondary; two series 3-ph bridges, interphase inductor

Mercury-arc rectifiers are usually **single-way connected** **(a, b, c, e)** but all connections suit silicon diodes. Connection **(e)** is common for high-current low voltage rectifier equipment. For higher voltages up to 2 kV, connection **(g)** can be used with two or more diodes in series for each path. In 6-pulse cases **(b, c, d)**, the transformer ratings are 1.2:1.2:1 with an inductor of rating 0.05 for **(b)**. The relative ratings for 12-pulse arrangements **(e, f, g)** are 1.3:1:1 with **(e)** and **(f)** requiring inductors respectively of rating 0.06 and 0.01.

Where harmonic limitation is essential, a **24-pulse system** may be used. This can be achieved by using two separate 12-pulse units with outputs in parallel and a 15° phase-shift obtained by a 7½° shift on each primary. Alternatively, four separate 6-pulse units may be used.

2. Traction-supply Transformers

High-voltage industrial-frequency railway traction systems (e.g. 25 kV, 50 Hz) are fed by single-phase track-side transformers spaced up to 40–50 km apart. Locomotive equipment must accommodate widely varying pick-up voltage resulting from supplying network, transformers and contact wire impedances, are designed to operate satisfactorily over the range 0.65 – 1.1pu of the nominal voltage.

Locomotives normally have **rectifier-fed DC motors**, so the substation load is likely to have a **distorted waveform** and to vary widely in amplitude (e.g. 0 to 2.5pu), giving **temperature rise** and **pulsating forces**.

The 1-ph substation transformers have fully insulated primary windings connected across traction lines. When the main supply network load is supplied with **off-load tap-changers** that may be set to account for **local supply regulation** and **traction conditions**.

Traction load is taken from a **large, interconnected network**, the effect of 3-ph unbalance and harmonic distortion can generally be accepted without significant interference on other loads and generating plants. But occasionally it may be necessary to get **harmonic filters**.

Service conditions require **short-circuit currents** to be limited by reactance. The leakage reactance (e.g. to be 0.1 p.u. for a unit of 10 MVA or above); and **surge overvoltages** up to 2.0 p.u to be withstood.

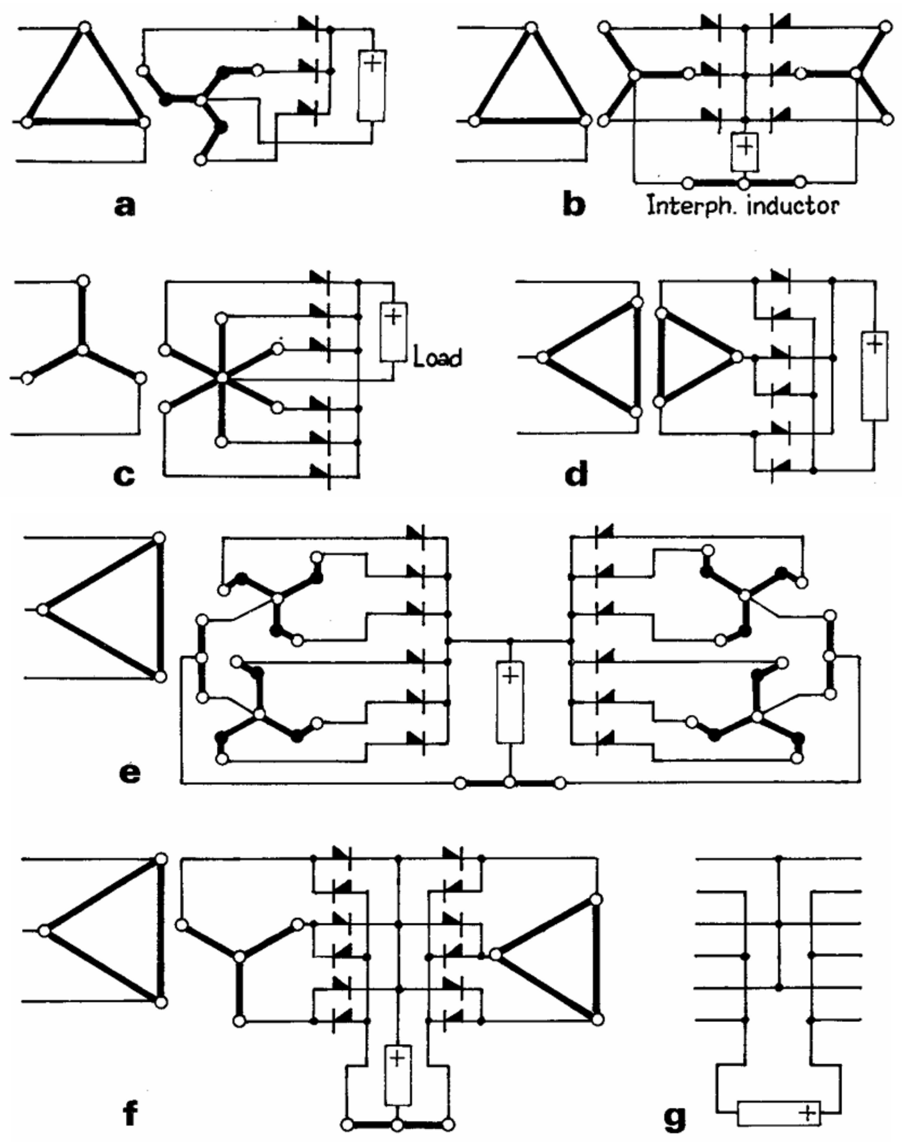


Figure 23: Rectifier Transformer Connections

15. Neutral Grounding

Purpose: Limits fault current by inserting high resistance between the neutral point of a transformer/generator and ground.

Delta Systems: Since delta configurations lack a neutral, an *artificial neutral* is created using a grounding transformer to enable HRG, as illustrated in **Fig. 24** and **Fig. 25**.

Benefits:

- Fault current limited to **2–10 A**, making fault location easier.
- Maintains **operational continuity** during single-phase earth faults.

Capacitance Discharge: In star-connected systems, line-to-ground capacitances can cause transient overvoltages during faults. NGRs must allow enough current to discharge these safely.

Time Ratings:

- Standard: 5–10 seconds
- Extended: Used in critical applications (e.g., mining, petroleum) to withstand longer fault durations.

Zigzag Transformer to Provide Earth Return Path

The zig-zag, or interconnected-star, earthing transformer is the usual system for earthing the 33 kV network. It presents a high impedance to balanced three-phase (positive-phase-sequence) voltages, but a **low impedance to in-phase (zero-sequence) voltages**. This is because the windings produce opposing fluxes in each core, under zero-sequence conditions. **Fig. 26** shows the current flow under a SLG fault.

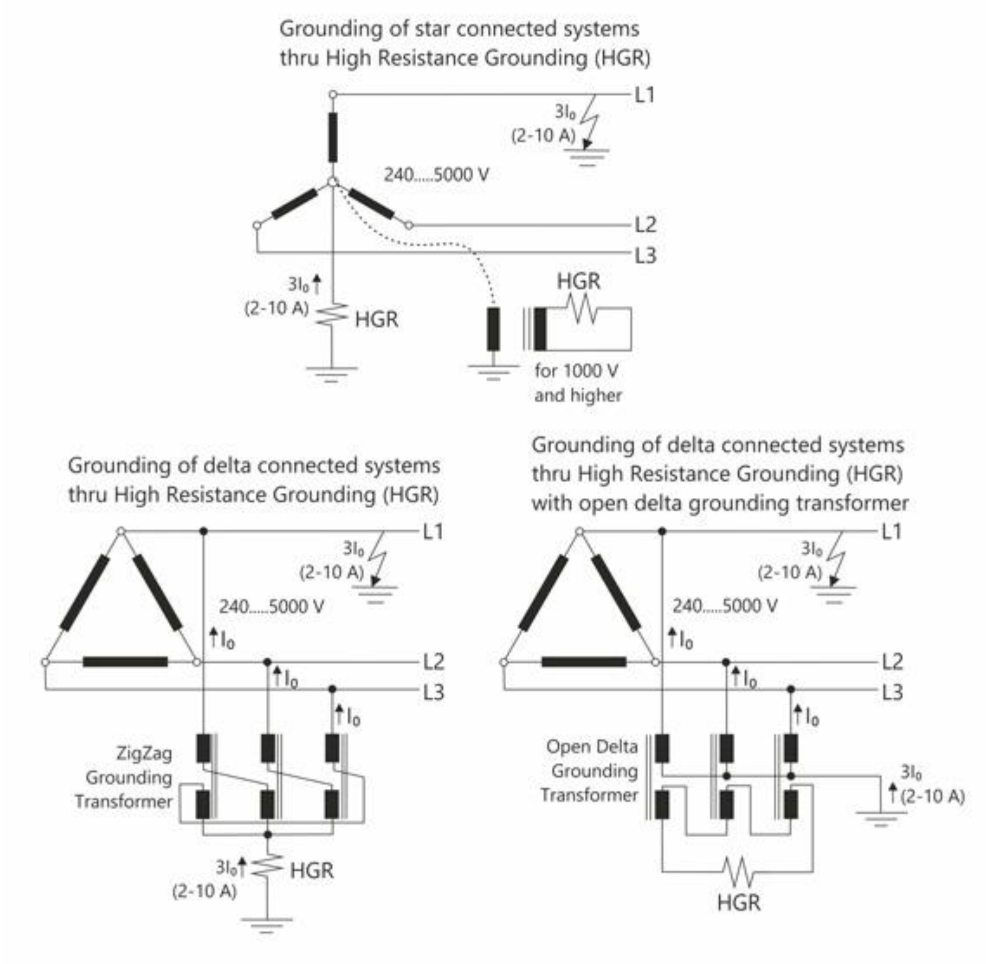


Figure 24: Different Applications on High Resistive Neutral Grounding Device

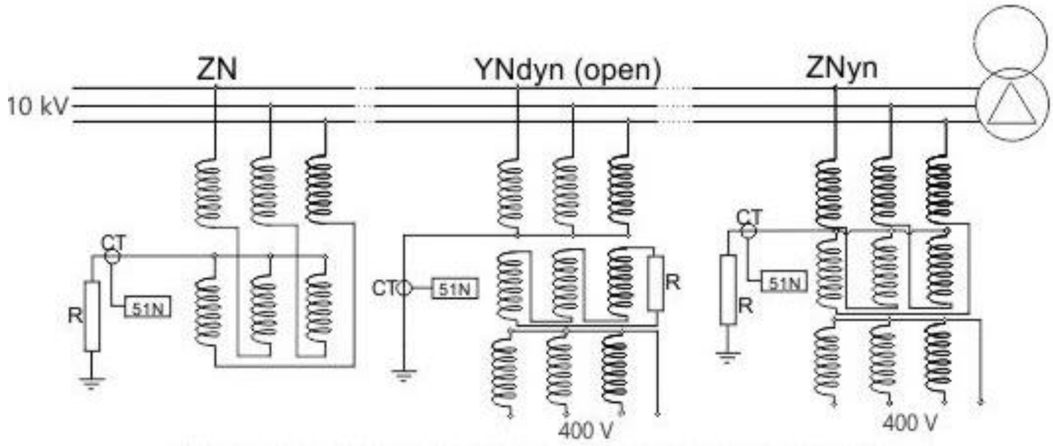


Figure 25: Examples of Artificial Neutral Point

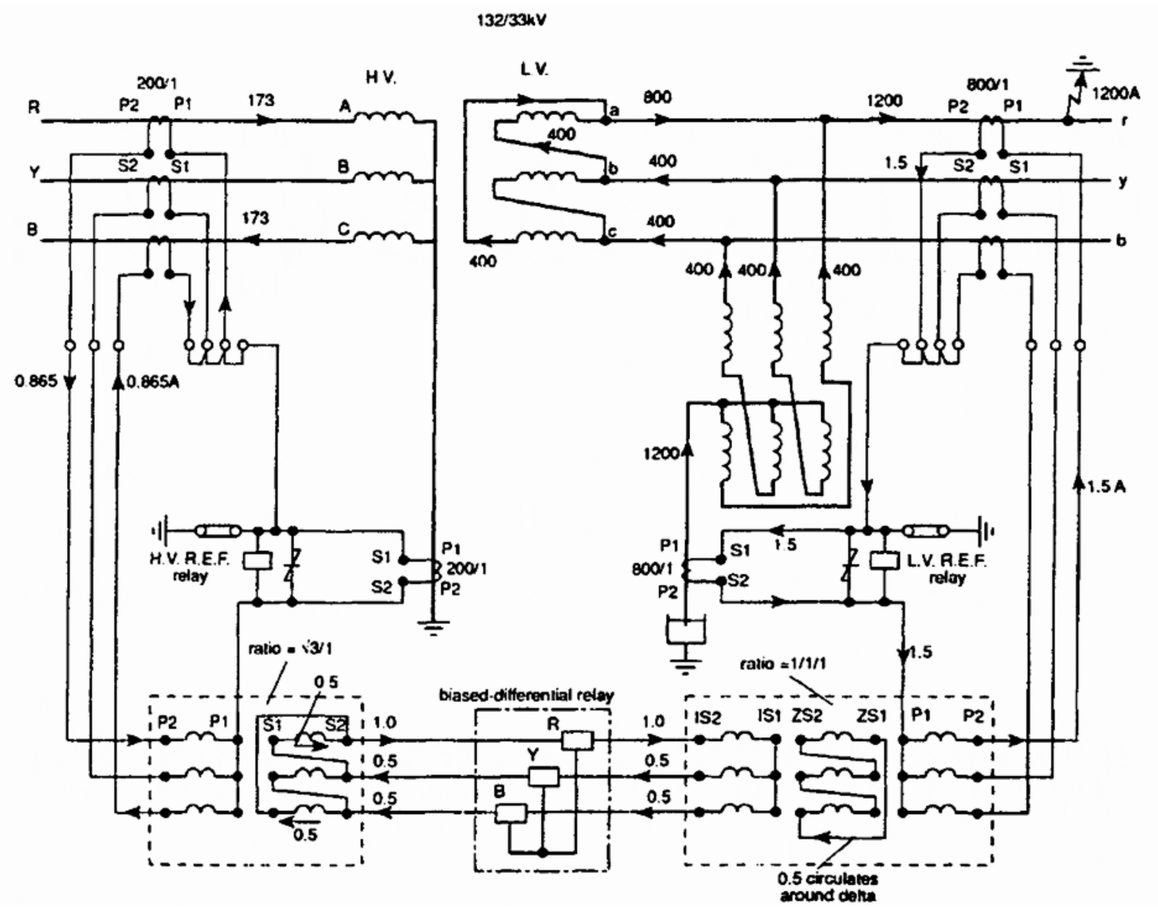


Figure 26: Protection on Star-Delta Transformer with Zigzag (Earthing) Transformer

THE END

Exam Questions regarding Theory of Transformers –

Spring 2013

a) Sketch the vector diagram of the three-phase Yz11 transformer. Hence, explain why the three-phase Yz11 transformer can offer phase and line output voltages without triplen harmonics.

b) Three three-phase transformers are available:
Transformer A: Y0, 132/11 kV, 500 kVA, (0.02 + j0.04) p.u.;
Transformer B: Dy1, 132/11 kV, 500 kVA, (0.02 + j0.04) p.u.;
Transformer C: Yz1, 132/11 kV, 250 kVA, (0.01 + j0.03) p.u.
Sketch the vector diagram of each transformer. Select two of them which can be operated satisfactorily in parallel and explain reason for your choice. Hence, calculate the apparent power loading and power factor of each of the selected two transformers when they are sharing an induction motor load of 500 kW at power factor 0.8 lagging.

Spring 2012

a) State the mandatory conditions and the preferable conditions for parallel operation of two three-phase transformers.

b) When two three-phase star/star-connected transformers, each having a negligible resistance and a leakage reactance of 0.05 Ω per phase referred to the secondary side, are fed from the same supply, their open-circuit secondary line voltages are 378 V and 382 V. The secondary windings of the two transformers are now connected in parallel to a balanced star-connected load of impedance 0.6 Ω per phase and of power factor 0.85 lagging.
Calculate the terminal line voltage of the load, and the complex power supplied by the transformer with the open-circuit secondary line voltage of 378 V.

Spring 2011

a) When three three-phase star/star-connected transformers T_{x1}, T_{x2}, T_{x3}, having internal impedances Z_1, Z_2 and Z_3 referred to the secondary side, are fed from the same three-phase voltage source, the corresponding open-circuit secondary phase voltages are E_1, E_2 and E_3, respectively. The secondary windings of all transformers are connected in parallel to a balanced star-connected load of impedance Z. Derive the expression of the terminal phase voltage V_L of the load.

b) Given the above transformer parameters:
- $Z_1 = (0.01 + j0.03)\ \Omega$
- $Z_2 = (0.01 + j0.04)\ \Omega$
- $Z_3 = (0.02 + j0.04)\ \Omega$
- $E_1 = 200$ V, $E_2 = 210$ V, $E_3 = 220$ V

They are connected in parallel to share a three-phase balanced star-connected load of 600 kVA at power factor 0.8 lagging.
Calculate V_L and hence the three-phase apparent power supplied by T_{x1}.

Spring 2009

a) Compare the merit and demerit between on-load tap-changing and off-load tap changing for power transformers. Describe with the aid of diagrams, the switching sequence of on-load tap changing.

b) When two three-phase delta/star-connected transformers, each having a negligible resistance and a leakage reactance of 0.05 Ω per phase referred to the secondary side, are fed in parallel from a supply of rated voltage, the corresponding open-circuit secondary line voltages are 377 V and 383 V, respectively. The secondary windings of the two transformers are now connected in parallel to a balanced star-connected load of impedance 0.5 Ω per phase and of power factor 0.8 lagging. Calculate the terminal line voltage of the load, and the complex power supplied by the transformer with the open-circuit secondary line voltage of 377 V.

Book of Example

Ideal Transformer Model and Transformer Formula

1. Find the turn-ratio (primary to secondary) of a 11000/400V, delta/star connected, three phase transformer.
2. A three-phase 50 Hz transformer core has a cross-section of 400 cm^2 (gross). If the flux density be limited to 1.2 Wb/m^2, find the number of turns per phase on high and LV side winding. The voltage ratio is 2200/220 V, the HV side being connected in star and LV side in delta. Consider stacking factor as 0.9.
3. A 50 Hz, three-phase core type transformer is to be built for an 11 kV /440 V ratio, connected in delta star. The cores are to have a square section and the coils are of circular. Taking an induced emf of 15 V per turn and maximum core flux density of about 1.1 T. Find the primary and secondary number of turns and cores cross-sectional area neglecting insulation thickness.
4. A three-phase, 50 Hz transformer of shell type has cross-sectional area of core as 400 cm^2. If the flux density is limited to 1.2 T, find the number of turns per phase on high voltage and low voltage side. The voltage ratio is 11000/400 V, the higher voltage side being connected in star and low voltage side in delta. Also determine the transformation ratio.

Connections, Open Circuit and Short Circuit Test, Regulation and Efficiency

5. A three-phase step down transformer is connected to 6600 V mains and takes a current of 24 amperes. Calculate the secondary line voltage, line current and output for the following connections (i) Delta-delta (ii) Star-star (iii) Delta-star (iv) Star-delta. The ratio of turns of per phase is 12. Neglect losses.
6. A three-phase, 50 Hz, core type transformer is required to be built for a 10000/500 V ratio, connected in star/mesh. The cores are to have a square section and the coils are to be circular. Taking an induced emf of about 15 V per turn and maximum core density of about 1.1T find the cross-sectional dimensions of the core and the number of turns per phase.
7. A three-phase transformer, rated at 1000 kVA, 11/3.3 kV has its primary star-connected and secondary delta connected. The actual resistances per phase of these windings are, primary 0.375 ohm, secondary 0.095 and the leakage reactances per phase are primary 9.5 ohm, secondary 2 ohm. Calculate the voltage at normal frequency which must be applied to the primary when the secondary terminals are short circuited. Calculate also the power under these conditions.
8. A 33/66 kV, 5 MVA, three-phase star-connected transformer with short circuited secondary passes full-load current with 7% primary potential difference and losses are 30 kW. With full potential difference on the primary and the secondary open circuited, the losses are 15 kW. What will be the efficiency of the transformer at full-load and 0.8 power factor?
9. A 2 MVA three-phase, 33/6.6 kV, delta/star transformer has a primary resistance of 8 ohm per phase and a secondary resistance of 0.08 ohm per phase. The percentage impedance is 7%. Calculate the secondary terminal voltage, regulation and efficiency at full load 0.75 power factor lagging when the iron losses are 15 kW.
10. The percentage impedance of a three-phase, 11000/400 V, 500 kVA, 50 Hz transformer is 4.5%. Its efficiency at 80% of full-load, unity power factor is 98.8%. Load power factor is now varied while the load current and the supply voltage are held constant at their rated values. Determine the load power factor at which the secondary terminal voltage is minimum.

Parallel Transformer

11. Two three-phase transformers each of 100 kVA are connected in parallel. One transformer has (per phase) resistance and reactance of 1% and 4% respectively and the other has (per-phase) resistance and reactance of 1.5% and 6% respectively. Calculate the load shared by each transformer and their pf when the total load to be shared is 120 kVA, 0.8 p.f. lagging.
12. Two 1000 kVA and 500 kVA, three-phase transformers are operating in parallel. The transformation ratio is same for both i.e., 6600/400, delta-star. The equivalent secondary impedances of the transformers are (0.001 + j0.003) ohm and (0.0028 + j0.005) ohm per phase respectively. Determine the load shared and pf of each transformer if the total load supplied by them is 1200 kVA at 0.866 pf lagging.

13. Two 400 kVA and 800 kVA transformers are connected in parallel. One of them (400 kVA transformer) has 1.5% resistive and 5% reactive drops whereas the other (800 kVA transformer) has 1% resistive and 4% reactive drops. The secondary voltage of each transformer is 400 V on load. Determine how they will share a load of 600 kVA at a pf of 0.8 lagging

Three Winding Transformer and Stabilizing Winding for Single Phase Load

14. The short circuit tests gave the following pu values of a 3-winding transformer. $Z12 = (0.012 + j0.064)$; $Z23 = (0.025 + j0.064)$; $Z31 = (0.016 + j0.12)$ The open circuit test on the primary gives the following values pu $Y0 = (0.02 – j\,0.05)$ The secondary is supplying full rated power at 0.8 pf lagging and tertiary is kept open. Find the impedance pu of each winding.
15. An 11 kV/400 V, three-phase, delta/star transformer supplies a load of 100 kW at unity pf between line R and neutral. It is also supplying a balanced load of 500 kW at 0.8 pf lagging. Determine the current magnitude in each primary winding and each input line. State assumptions if any and draw the relevant diagram
16. A 3300/400/110 V star-star-delta transformer takes a magnetising current of 6A and a balanced load of 500 kVA at 0.8 pf lagging and 200 kVA at 0.6 pf leading on the tertiary. Determine the primary current and its p.f
17. A star/star/delta connected transformer has secondary load of 40A at 0.8 pf lagging and tertiary has load of 30A at 0.71 pf lagging. The ratio of turns is 10 : 2 : 1 for primary, secondary and tertiary respectively. Calculate the primary current and pf.

Vee-Vee Connection and Scott Transformer

18. A balanced three-phase load of 1000 kW and 0.8 p.f. lagging is supplied by Vee-connected transformers. Calculate the line and phase currents and the power factor at which each transformer is working. The working voltage is 3.3 kV.
19. Two identical single phase transformers are connected in V-V-connection across three-phase mains and deliver a balanced load of 3048 kW at 11 kV, 0.8 p.f. lagging. Calculate the line and phase currents and the pf at which each transformer is working. The V-V-connection is converted to mesh/mesh by the addition of an identical unit. Calculate the additional load of same p.f. that can now be supplied for the same temp. rise. Also calculate the phase and line currents.
20. Two-transformers are connected in open delta and supply a balanced three-phase load of 300 kW at 400 volt and a p.f. of 0.866, determine: (a) the secondary line current. (b) the kVA load on each transformer (c) the power delivered by the individual transformers. If a third transformer having the same rating i.e., 200 kVA is added to form a delta bank, what will be the total load that can be supplied
21. A three-phase 150 kW balanced load at 1000V, 0.866 p.f. lagging is supplied from 2000 volt, three phase main through (i) three single-phase transformers connected in (i) delta-delta and then (ii) in Vee-Vee. Find the current in the windings of each transformer and the p.f. at which they operate in each case.
22. Three 1100/110 V transformers are connected in delta-delta and supply a lighting load of 120 kW. If one of these transformers is damaged and hence removed for repairs, what currents will be flowing in each transformer when (i) the three transformers were in service (ii) the two transformers are in service.
23. Three 1100/110 volt transformers are, connected in delta/delta. This bank of transformers is supplying a lighting load of 100 kW. One of these transformers is damaged and removed for repairs. (i) What currents were flowing in each transformer when the three transformers were in service. (ii) What current flows in each of the two transformers after third has been removed. (iii) What is the kVA output of each transformer if the two transformers, connected in open-delta, carry full load with normal heating.
24. The primary and secondary windings of two transformers each rated 220 kVA, 11/2 kV and 50 Hz are connected in open delta. Find (i) the kVA load that can be supplied from this connection; (ii) currents on HV side if a delta connected three phase load of 250 kVA, 0.8 pf (lag) 2 kV is connected to the LV side of the connections.
25. Two single phase furnaces A and B are supplied at 100 V by means of Scott-connected transformers from a three-phase, 6 kV system. Furnace A is supplied from the teaser transformer. Calculate the line

currents on the three phase side when (i) each furnace takes 600 kW at p.f. 0.8 (lag). (ii) furnace A takes 500 kW at unity power factor and furnace B 600 kW at 0.8 pf (lag).

26. Two one-phase furnaces A and B are supplied at 80 V by means of a Scott-connected transformer combination from a three-phase 6600 V system. Calculate the line currents on the three-phase side when the furnace A takes 500 kW at unity p.f. and B takes 800 kW at 0.7 p.f. lagging. Draw the corresponding vector diagrams.
27. Two single-phase electric furnaces are supplied power at 80 volt from a three-phase, 11 kV system by means of two single-phase Scott-connected transformers, with similar secondary windings. When the load on one transformer is 500 kW and on the other is 800 kW, what current will flow in each of the three-phase lines at unity p.f.
28. It is desired to transform 2000 V, 400 kVA three phase power to two-phase power at 500 V by Scott connected transformers. Determine the voltage and current ratings of both primary and secondary of each transformer. Neglect the transformer no-load currents
29. A balanced three-phase, 100 kW load at 400 V and 0.8 pf lagging is to be obtained from a balanced two phase 1100 V lines. Determine the kVA rating of each unit of the Scott-connected transformer.
30. Two single phase Scott-connected transformers supply a three-phase four-wire 50 Hz distribution system with 400 V between lines. The hv windings are connected to a two-phase 6600 V (per phase) system. The core area is 200 cm2, while the maximum allowable flux density is 1.2 T. Determine the number of turns on each winding and the point to be tapped for the neutral wire on the three-phase side.

Review Questions

1. A single unit of three-phase transformer is preferred over three units of one-phase transformers to transfer power in three-phase circuits, why?
2. LV winding is always placed nearer the core, why?
3. Parallel operation of the transformers in the power system is necessary, why?
4. Tap-changers are usually employed on the HV side, why?
5. No-load tap changers are only operated at off load, why?
6. A reactor is employed with on-load tap changer, why?
7. In power transformer, breather is necessary, why?
8. What is the importance of cooling of transformer?
9. Write a short note on a star-Delta transformer.
10. Sketch the three possible resistive grounding methods for transformers, e.g. through neutral earthing transformer, through zigzag transformer and through open delta as tertiary.
11. Explain the construction of a three-phase (i) core type and (ii) shell type transformers with the help of neat sketches.
12. State and explain the conditions necessary for parallel operation of two three-phase transformers.
13. Explain with the help of a phasor diagram the load shared by each transformer when two are connected in parallel having same voltage ratios but different impedance triangles.
14. What is a tap-changer? Where and how they are placed with the transformers?
15. Explain the working of an on-load tap-changer with the help of neat schematic diagram.
16. Compare delta and open delta connections.
17. Why Scott connections are also known as T-connections?
18. Why a tertiary winding is also called as auxiliary winding and stabilising winding?
19. Star-star connected transformers (without ground) are rarely used, why?

THE END